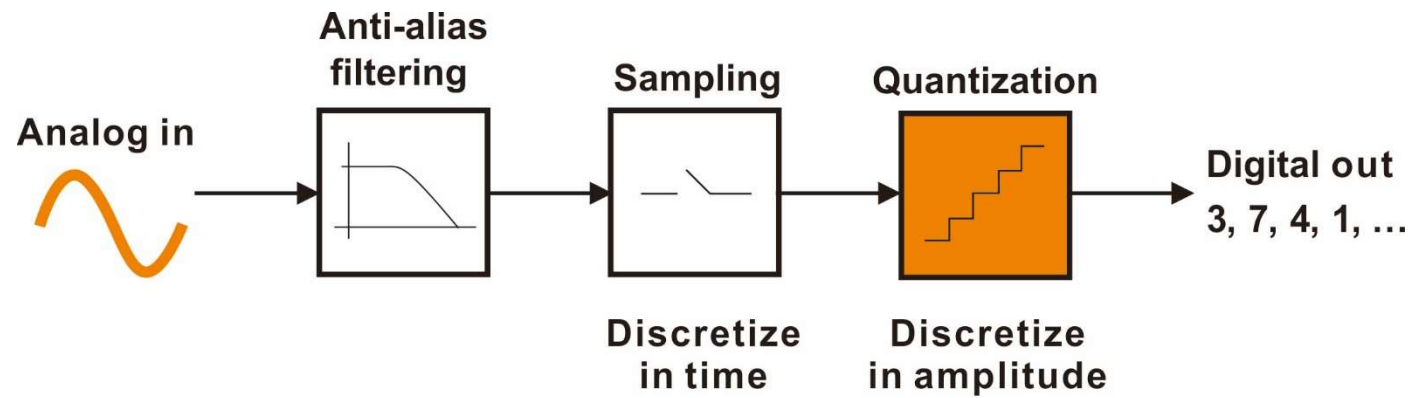


Voltage Comparators

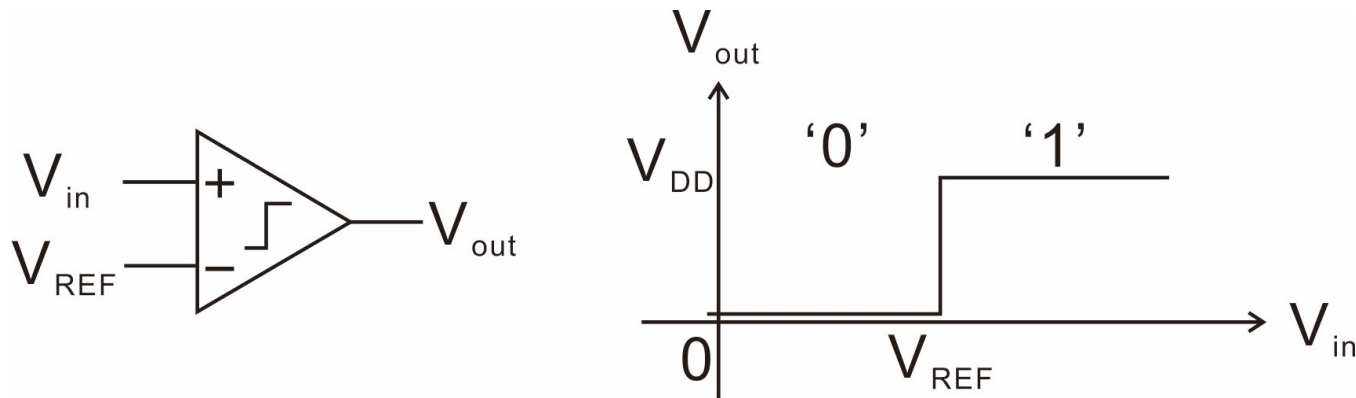
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ADC Big Picture



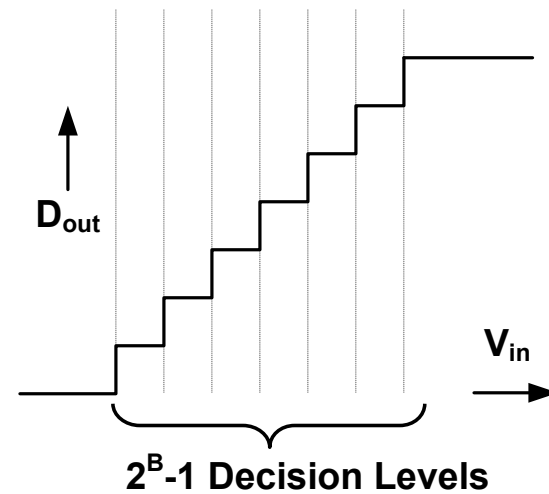
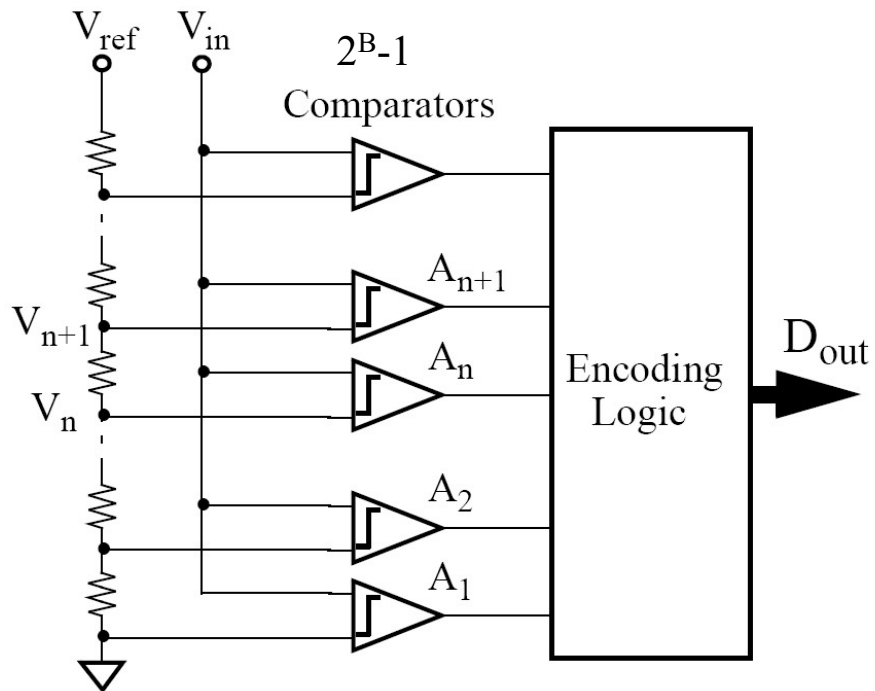
- How to quantize a voltage signal?
 - Use voltage comparators to make decisions

Ideal Voltage Comparator



- The comparator acts as a 1-bit ADC. It converts an “analog” input into a “digital” output.
- How to get more bits?
 - Use an array of comparators

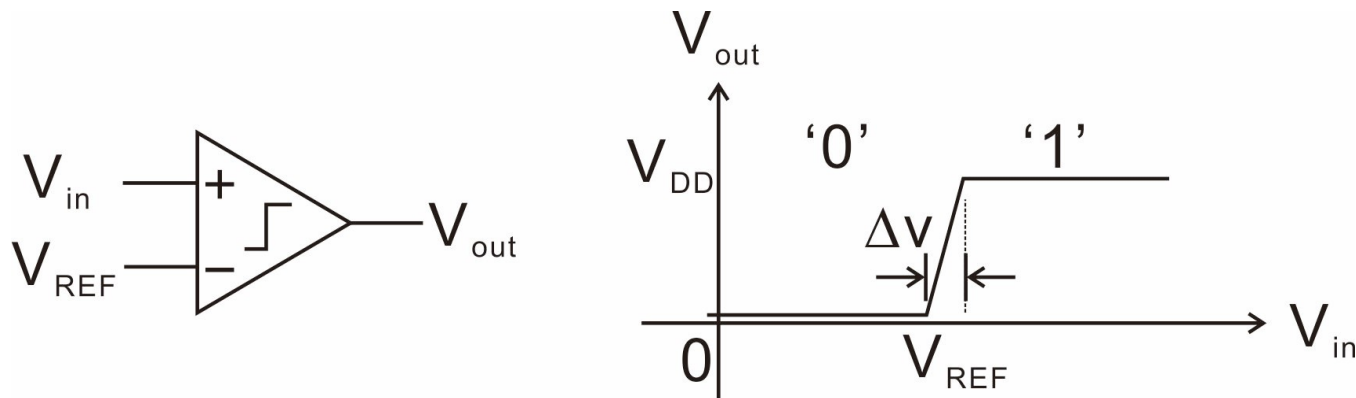
Flash ADC Basic Concept



Design Considerations

- Accuracy
 - Comparator resolution
 - Comparator offset
- Speed
 - Comparator delay
 - Comparator recovery time
- Power dissipation
- Other properties
 - Common-mode rejection
 - Input capacitance and linearity of input capacitance
 - Kickback noise

Gain Requirements



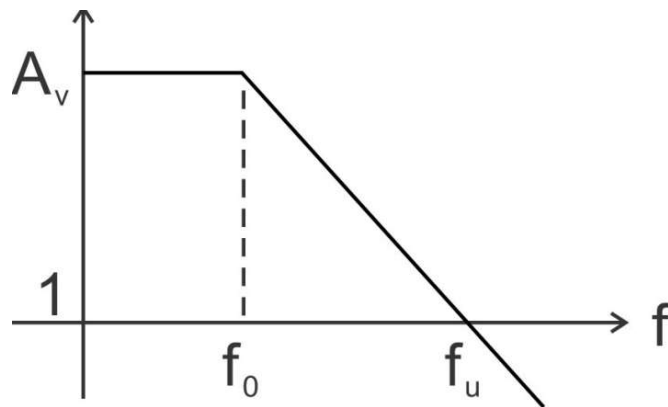
- Comparator output is 'digitally' unknown for $V_{REF} - \Delta/2 < V_{in} < V_{REF} + \Delta/2$
- We need comparator with a large effective gain A_V
- Assuming $\Delta V = 1\text{mV}$ and $V_{DD} = 1.8\text{V}$, we have:

$$A_V = \frac{1.8\text{V}}{1\text{mV}} = 1800$$

- How can we achieve this high gain?

How about Using an Op-Amp?

- Opamp can achieve high gain, but its speed is a problem.
- For an opamp, $A_v = g_m R$, $\omega_0 = 1/RC \rightarrow A_v \omega_0 = g_m/C = I_d/C * (g_m/I_d)$
 - g_m/I_d depends on biasing, but usually around $10 \sim 20 \text{ V}^{-1}$
 - For the same I_d (power), a direct trade-off between A_v and f_0



E.g., $f_u = A_v f_0 = 1\text{GHz}$ and $A_v = 1800$

$$\Rightarrow f_0 = \frac{f_u}{A_v} \approx \frac{1\text{GHz}}{1800} \approx 500 \text{ kHz}$$

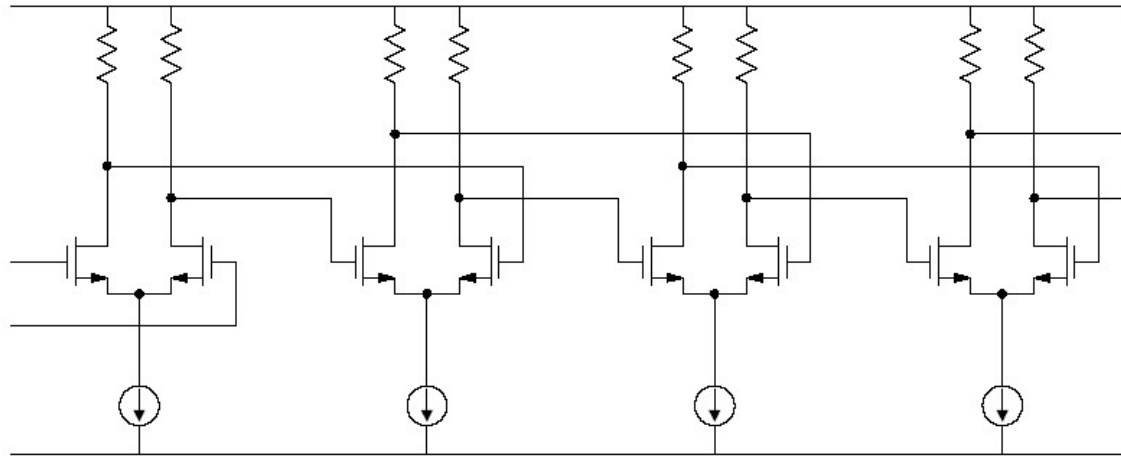
Too slow!

- We can trade power for speed: 2X power $\rightarrow$ 2X speed
- Is there a smarter approach?

Cascade 2 Amplifiers

- Analysis
 - 2 amplifier $\rightarrow$ 2X power
 - Total gain $A_v \rightarrow$ each amplifier with gain of $A_v^{0.5}$ and BW of $A_v^{0.5}f_0$
 - Thus, the overall bandwidth $\approx A_v^{0.5}f_0 \gg 2f_0$
 - Much favorable tradeoff!
- Can cascade even more

Cascade of Amplifiers



In each stage: $A_0 = A_V^{1/N}$ $\omega_0 = \frac{\omega_u}{A_0}$

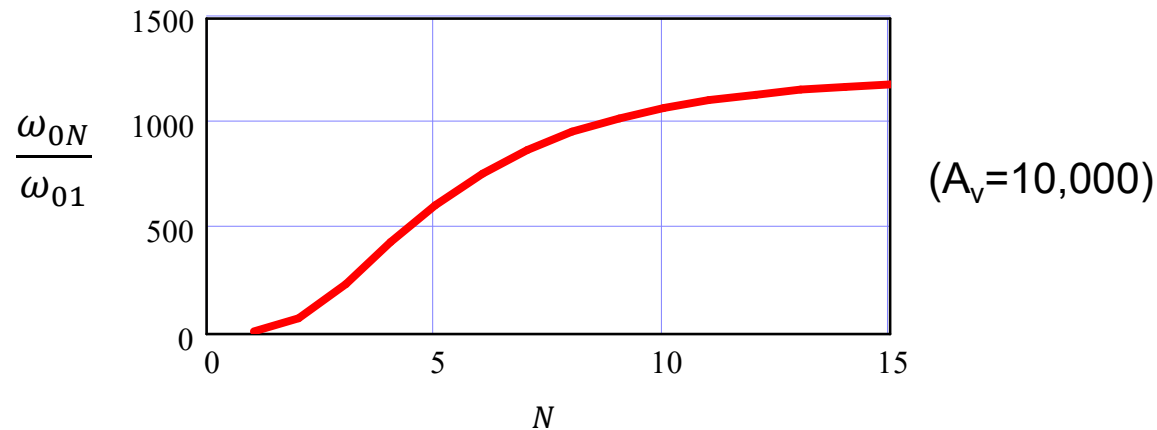
Overall: $A_V(s) = \frac{A_V}{(1 + s/\omega_0)^N}$ $\omega_{0N} = \frac{\omega_u}{A_0} \sqrt{2^{1/N} - 1}$

A_V : Total gain requirement

ω_{0N} : 3dB bandwidth of N stage realization

Bandwidth Perspective

$$\omega_{0N} = \omega_u \frac{\sqrt{2^{1/N} - 1}}{A_V^{1/N}}$$

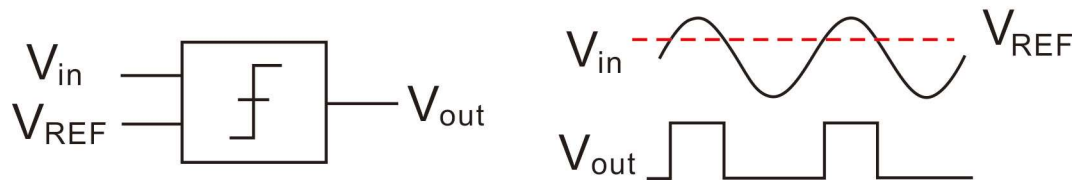


- As N increases, the return in BW diminishes and the power goes up. We do not want too large N .
- We also do not prefer a large N from the delay perspective.

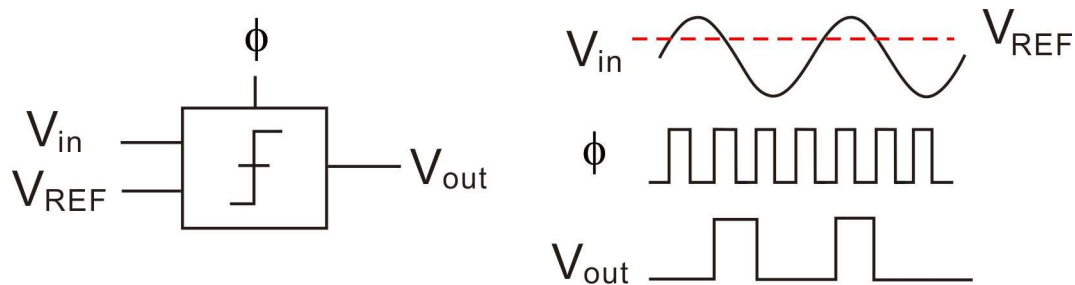
Bandwidth vs. Delay

- What we care most is the comparator delay.
- Comparators in ADCs are usually clocked.
 - Comparison starts when $\phi = 1$; reset when $\phi = 0$
- Analysis: the input is equivalent to a step

Continuous-time comparison



Clocked comparison



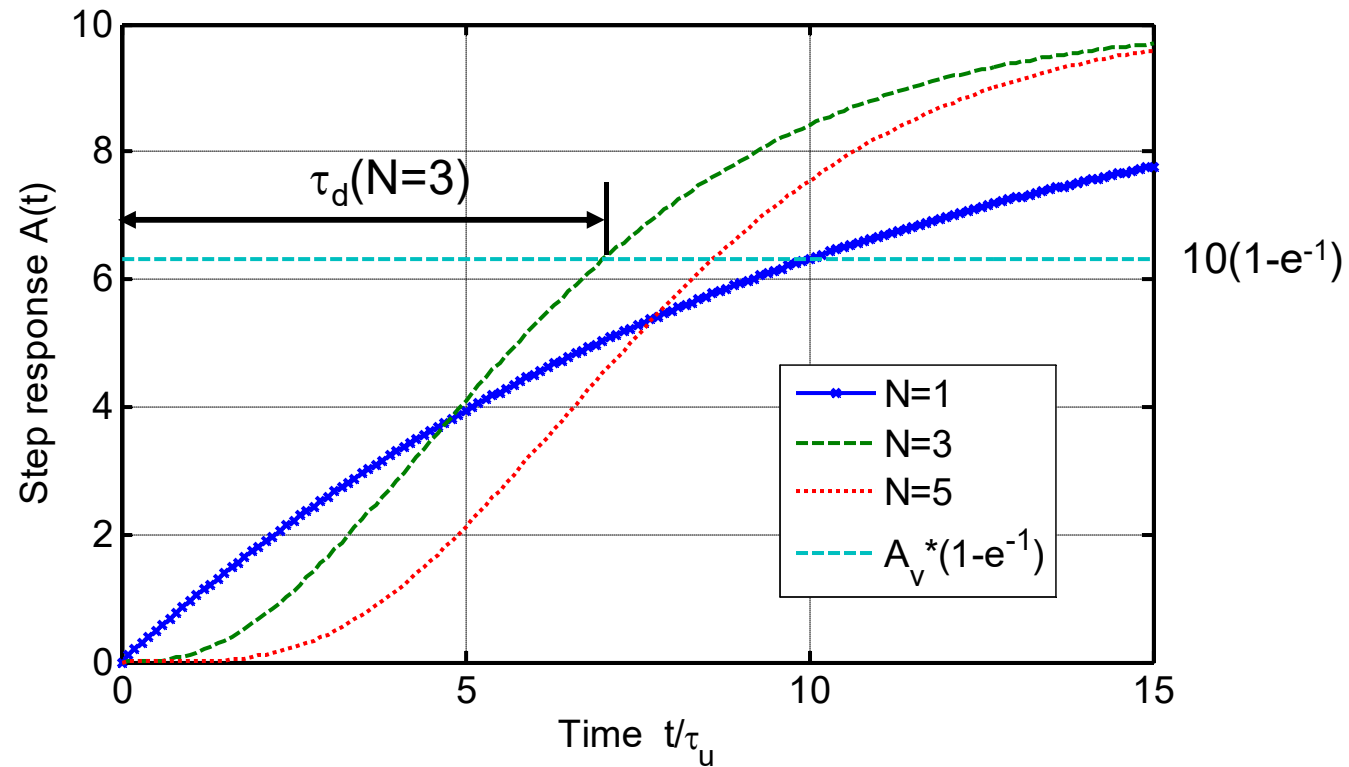
Step Response (1)

- Calculate the comparator delay for a step input:

$$V_{out}(s) = V_{in}(s)A(s) = \frac{V_{step}}{s} \frac{A_V}{(1 + s \cdot \tau_u A_V^{1/N})^N} \quad \tau_u = \frac{1}{\omega_u}$$

$$V_{out}(t) = V_{step} A_V \underbrace{\left(1 - e^{\frac{-t}{\tau_u A_V^{1/N}}} \sum_{k=0}^{N-1} \frac{\left(\frac{t}{\tau_u A_V^{1/N}} \right)^k}{k!} \right)}_{A(t)}$$

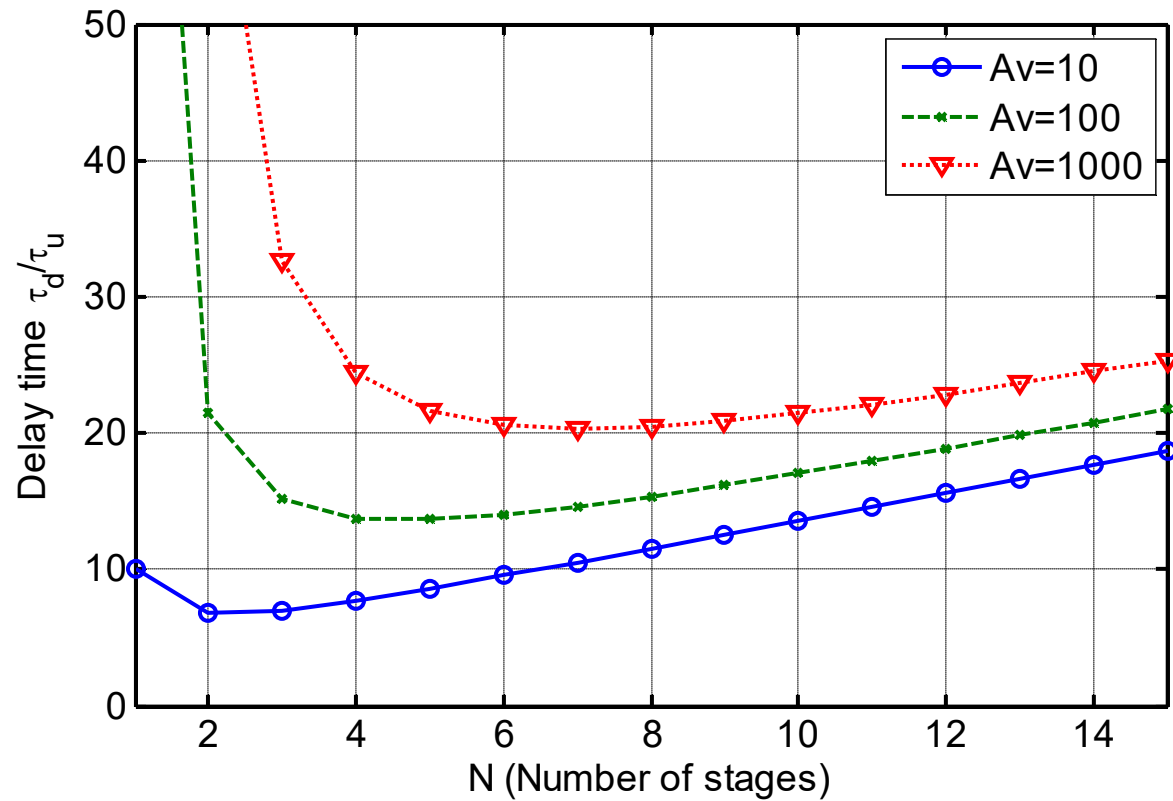
Step Response (2)



- 3-stage amplifier wins for $A_v=10$!

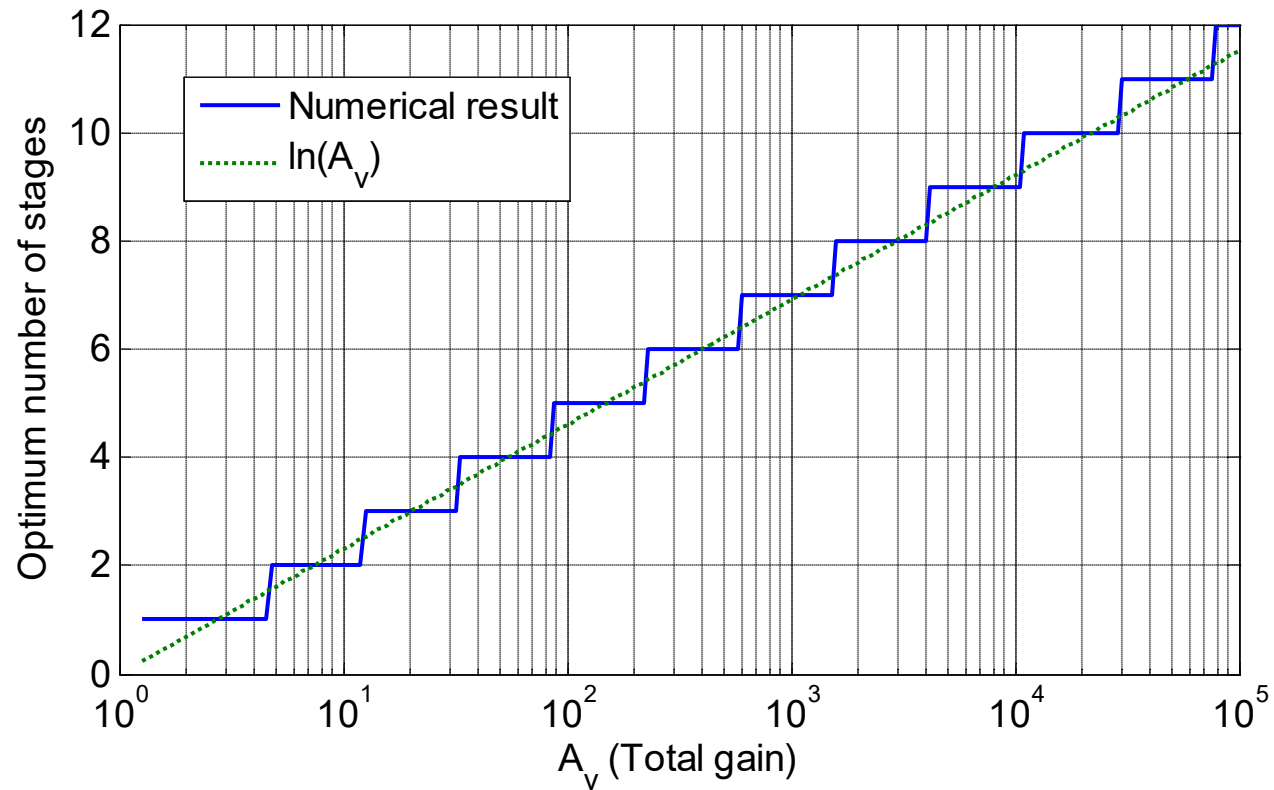
Delay vs. Number of Stages

$$A(\tau_d) = A_V(1 - e^{-1})$$
$$\Rightarrow \tau_d$$



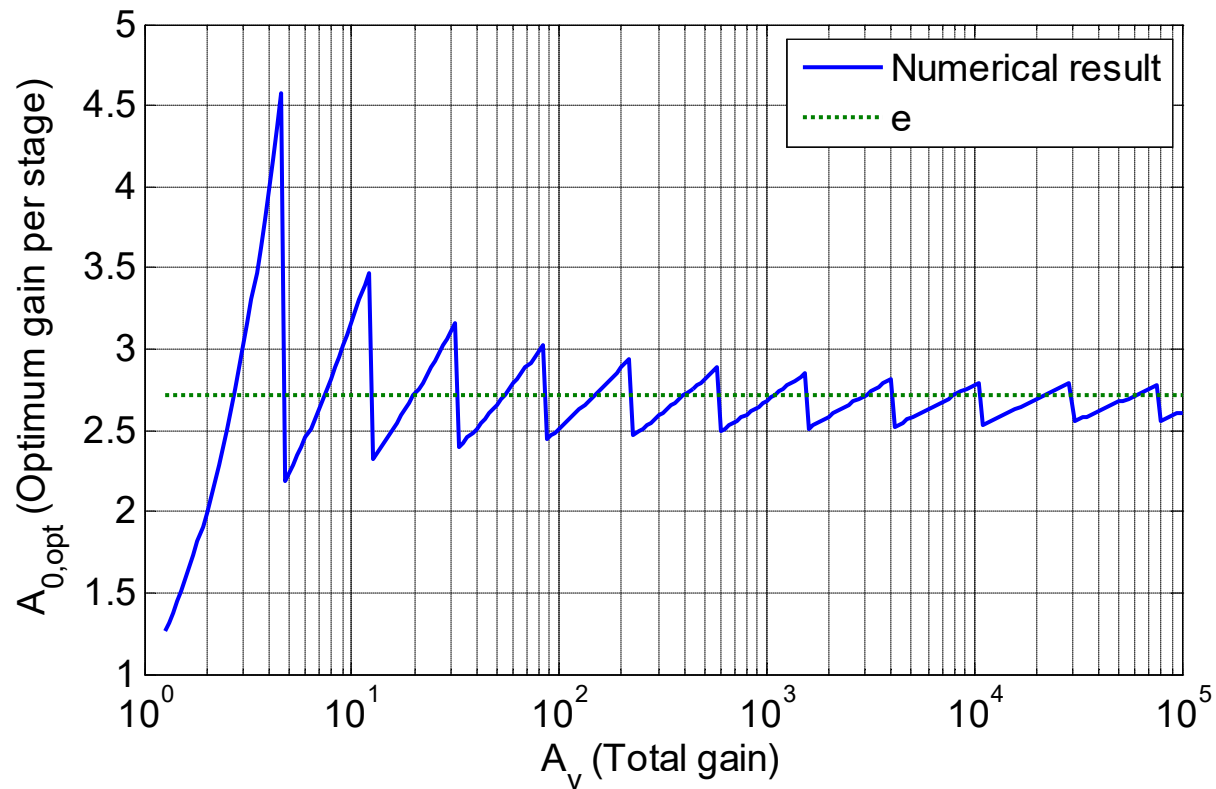
- The minima are shallow

Optimum Number of Stages



$$N_{opt} \approx \ln(A_v)$$

Optimum Gain per Stage

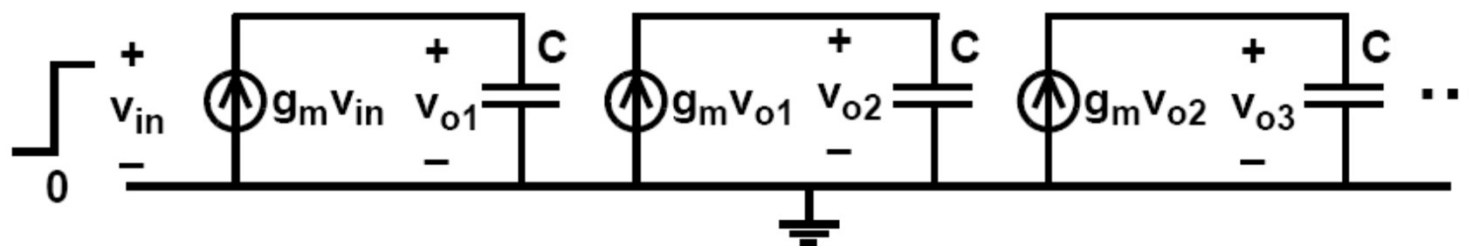


$$N_{opt} \approx \ln(A_V) \quad e^{N_{opt}} = A_V = A_{0,opt}^{N_{opt}} \quad \Rightarrow \quad A_{0,opt} \approx e$$

Can We Further Increase Speed?

- Cascade multi-stage integrators
- Regenerative latch using positive feedback
 - E.g. cross coupled inverters

Cascade of Integrators

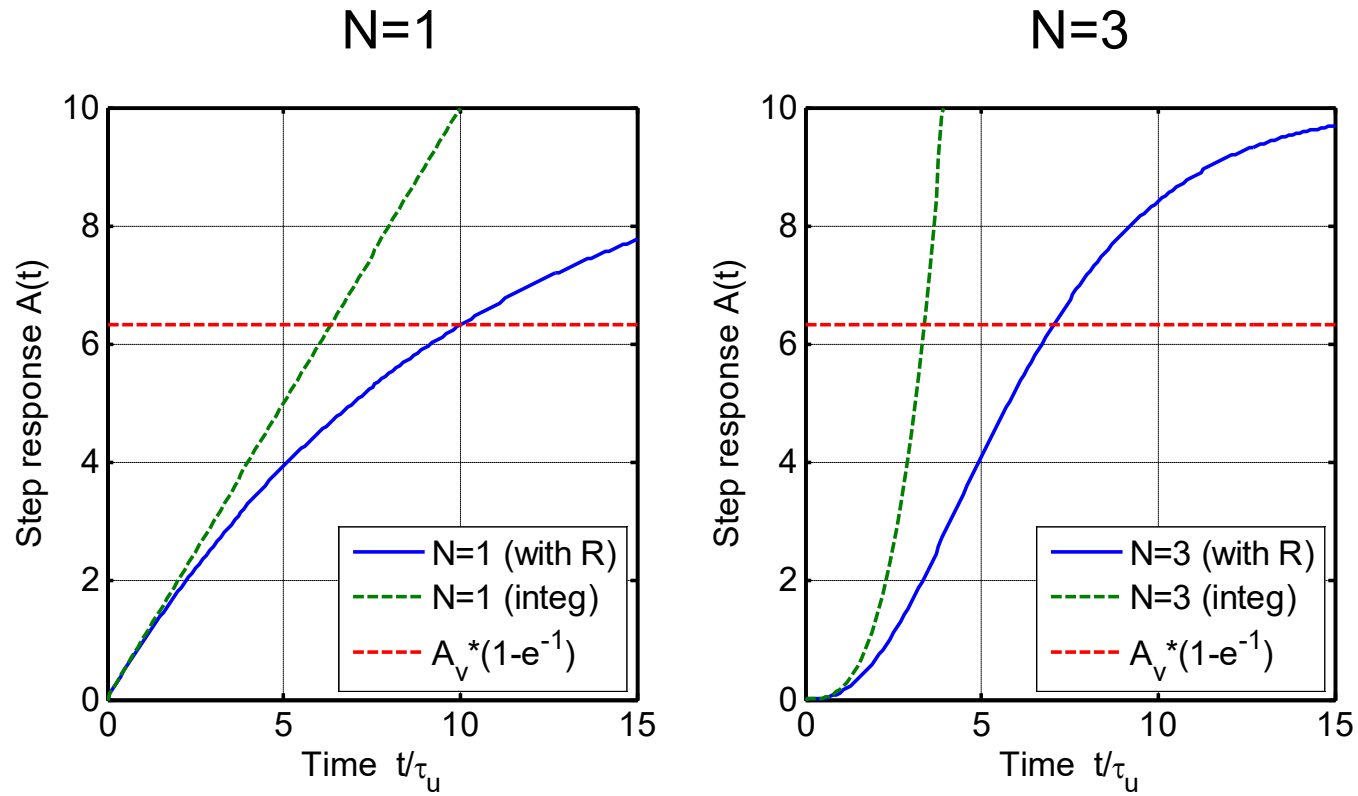


- Intuition
 - Load resistors (in cascade of open loop amplifiers) shunt signal current away from load capacitance $\rightarrow$ this slows down the amplification
 - Let us use all signal currents to charge the load capacitor
- Step response analysis

$$v_{o1} = \frac{g_m}{sC} v_{in} = \frac{\omega_u}{s} v_{in} \Rightarrow v_{oN} = \frac{\omega_u^N}{s^N} v_{in}$$

$$V_{out}(s) = \frac{V_{step}}{s} \frac{\omega_u^N}{s^N} \quad V_{out}(t) = V_{step} \omega_u^N \frac{t^N}{N!} \quad A(\tau_d) = \omega_u^N \frac{t^N}{N!}$$

Simulation Results



Analysis

- Cascade of integrators is faster than cascade of resistively loaded stages
- Integrator requires reset, but not a problem due to clocked comparison
- Delay time

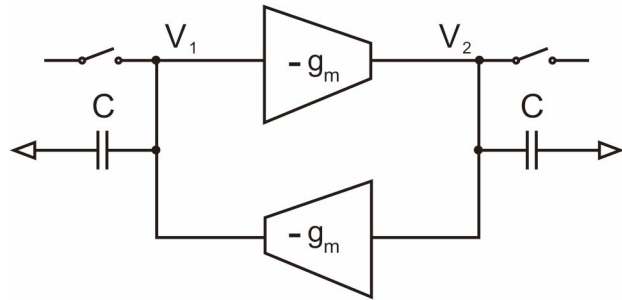
$$\tau_d = \tau_u [N! A_v]^{1/N}$$

- Optimum number of stages approximately given by

$$N_{\text{opt}} = 1.1 \ln[A(\tau_d)] + 0.79 \quad [\text{Wu, JSSC 12/1988}]$$

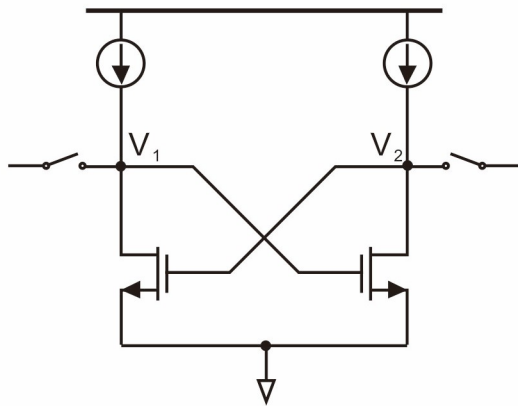
- Effective gain per stage is still close to $e \approx 2.7183$
- Can we further increase the speed but without requiring many stages?

Use Positive Feedback – Regenerative Latch



$t < 0$ setup initial condition $v_{10} - v_{20} = v_{d0}$

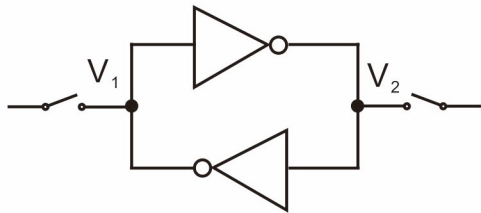
$t \geq 0$ enable positive feedback



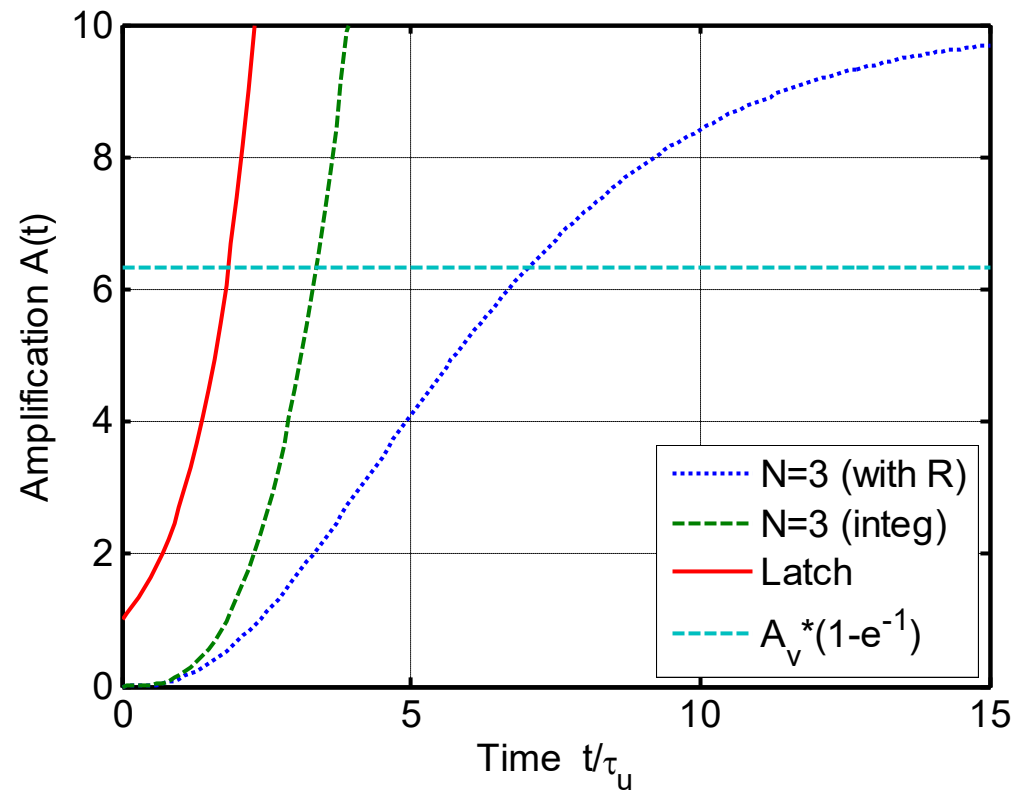
$$\begin{cases} \frac{dv_1}{dt} = \frac{i_1(t)}{C} = \frac{-g_m v_2(t)}{C} \\ \frac{dv_2}{dt} = \frac{i_2(t)}{C} = \frac{-g_m v_1(t)}{C} \end{cases}$$

$$\Rightarrow v_1(t) - v_2(t) = v_d(t) = v_{d0} e^{t/\tau_u}$$

$$\Rightarrow A(t) = \frac{v_d(t)}{v_{d0}} = e^{t/\tau_u} \quad \tau_u = \frac{C}{g_m}$$



Comparison



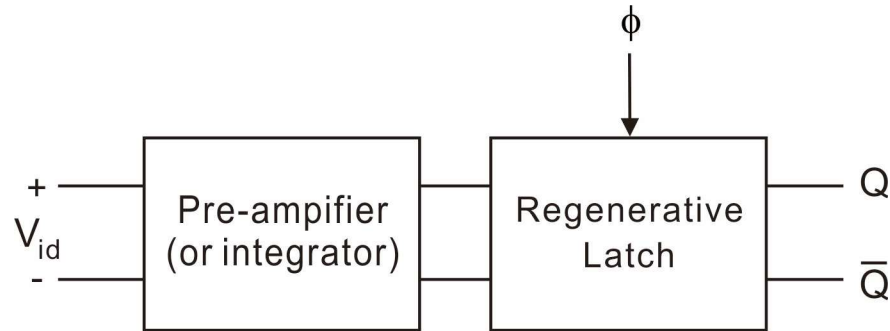
- Exponential wins over polynomial!
- Latch is much faster than cascade of amplifiers/integrators

Latch "Gain"

$A(\tau_d)$	τ_d/τ_u
10	2.3
100	4.6
1,000	6.9
10,000	9.2

- Trade exponential gain (A) with linear delay (τ_d)
- Better than amplifier and integrator

Standard Comparator Architecture



- Why bother using pre-amplification?
 - Offset
 - Use pre-amplification to lower input referred offset
 - Hard to build latches with offset $< 10\sim 100\text{mV}$
 - Common mode rejection
 - Comparator offset can be input dependent
 - Attenuate "kickback noise"
 - Metastability

Metastability (1)

- Consider minimum initial latch input voltage needed to regenerate to V_{DD} within maximum available time T_{max}

$$v_{d0min} = \frac{V_{DD}}{e^{T_{max}/\tau_u}}$$

- Minimum-required pre-amplifier input

$$v_{id0min} = \frac{1}{A_V} \frac{V_{DD}}{e^{T_{max}/\tau_u}}$$

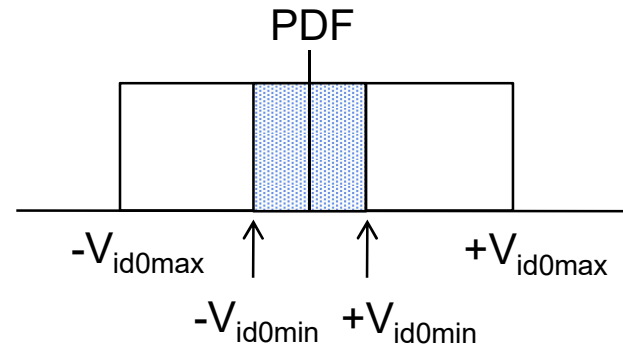
- Probability of seeing a metastable output

$$P(\text{Error}) = P(V_{id0} < V_{id0min})$$

Metastability (2)

- Assuming a uniform input signal distribution over some range, we have

$$P(\text{Error}) = \frac{V_{id0min}}{V_{id0max}}$$



- In a flash ADC, only 1 comparator can be metastable. The effective input range V_{id0max} over which the signal is distributed is thus $\Delta/2$

$$P(\text{Error}) = \frac{\frac{1}{A_V} V_{DD} e^{-T_{max}/\tau_u}}{\frac{\Delta}{2}} = \frac{2^{B+1} V_{DD}}{A_V V_{FS}} e^{-T_{max}/\tau_u}$$

Metastability (3)

- Example: 6-bit, 1GHz Flash ADC, $T_{\max}=T_s/2=0.5\text{ns}$, $\tau_u=1/(2\pi\cdot 10\text{GHz})=16\text{ps}$, $A_v=3$, $V_{\text{FS}}=V_{\text{DD}}$

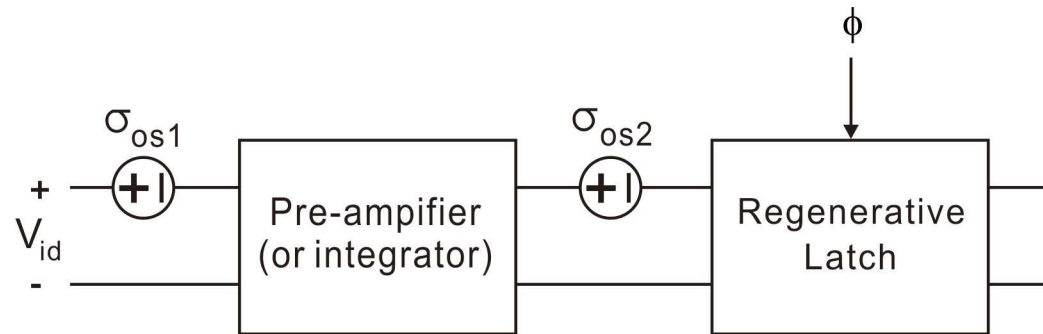
$$P(\text{Error}) = \frac{2^{6+1}}{3} e^{-500/16} \approx 10^{-1}$$

- Mean time to failure (MTF)

$$MTF = \frac{1}{P(\text{Error})f_s} = \frac{1}{10^{-12} \times 1 \times 10^9} \text{sec} = 1000\text{sec} \approx 17\text{min}$$

- We can enlarge MTF by
 - Reducing speed (larger $T_{\max}/\tau_u$)
 - Exponential dependence
 - Increasing pre-amplifier gain A_v
 - Linear dependence

Comparator Offset



$$\sigma_{os}^2 = \sigma_{os1}^2 + \frac{1}{A_V^2} \sigma_{os2}^2$$

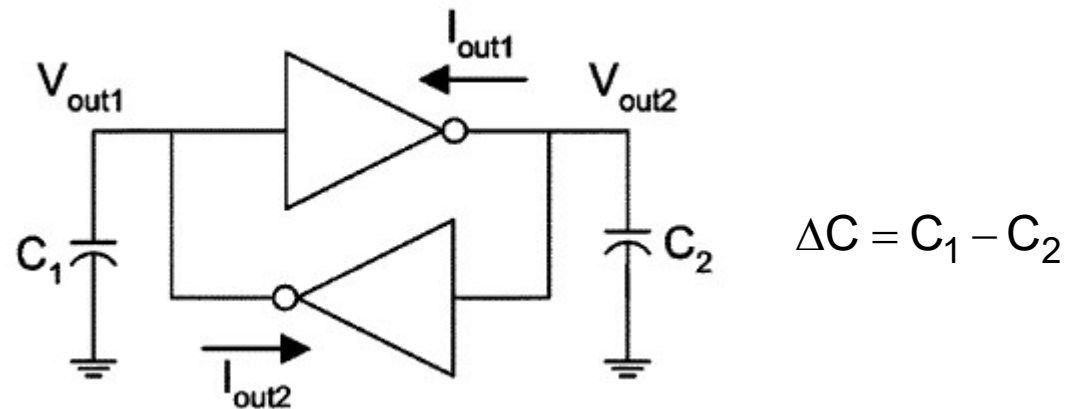
- Example: $\sigma_{vos1}=1\text{mV}$, $\sigma_{vos2}=10\text{mV}$, $A_V=10$

$$\sigma_{vos} = \sqrt{(1\text{mV})^2 + \frac{1}{10^2} (10\text{mV})^2} = 1.4\text{mV}$$

Causes for Comparator Offset

- V_{th} mismatch
- Device dimension mismatch
- Load capacitance mismatch
 - Load both sides even if only one side output is needed
 - Be careful about wiring

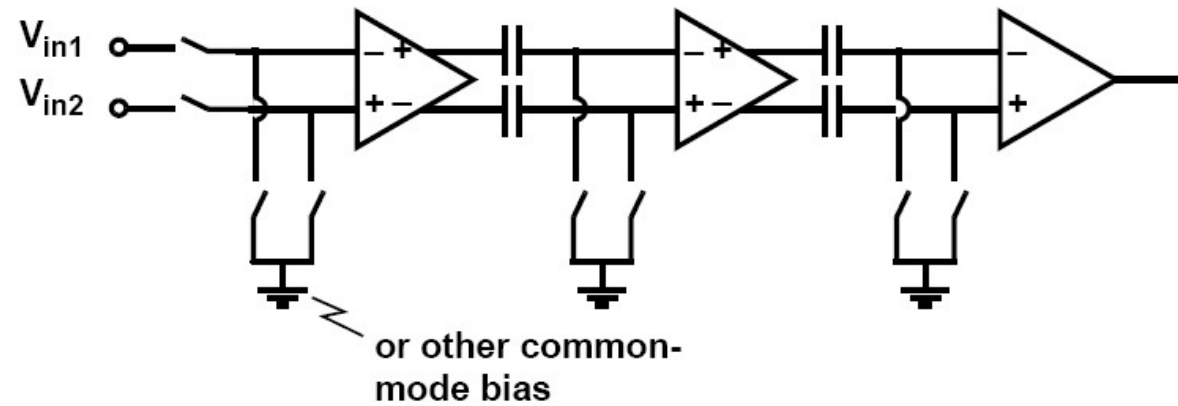
Load Capacitance Mismatch



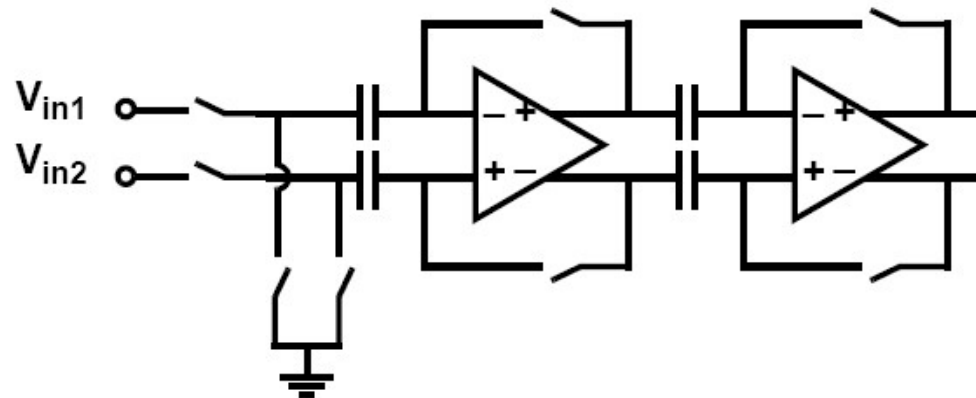
- Can show $V_{os} \cong 0.5(V_{(t=0)} - V_S)\Delta C/C$; V_S = inverter switching voltage
 - Nikoozadeh & Murmann, IEEE TCAS II, Dec. 2006.
- Example
 - $0.5 \cdot 10\text{fF}/100\text{fF} \cdot (1\text{V} - 0.5\text{V}) = 25\text{mV}$ (!)
- Need to match the output load.

Preamplifier Offset Cancellation

Output Series Cancellation

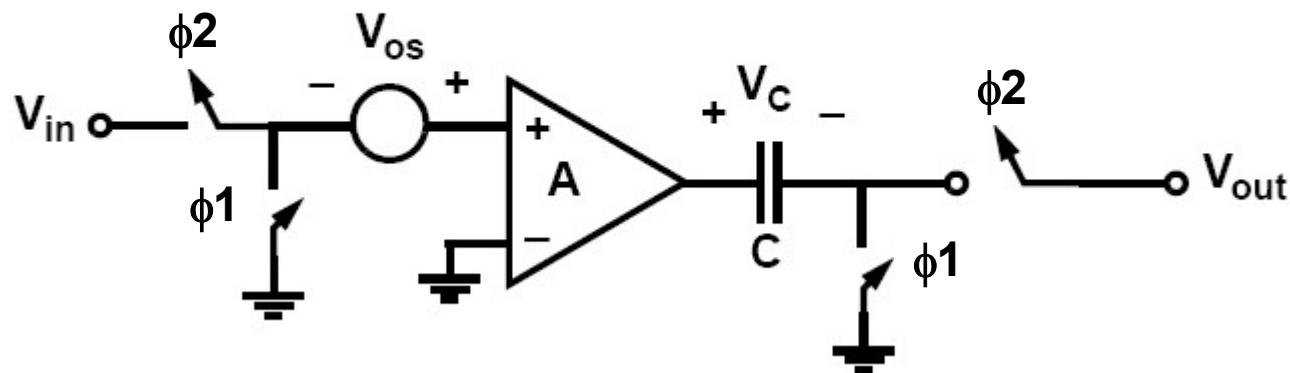


Input Series Cancellation



Output Series Cancellation

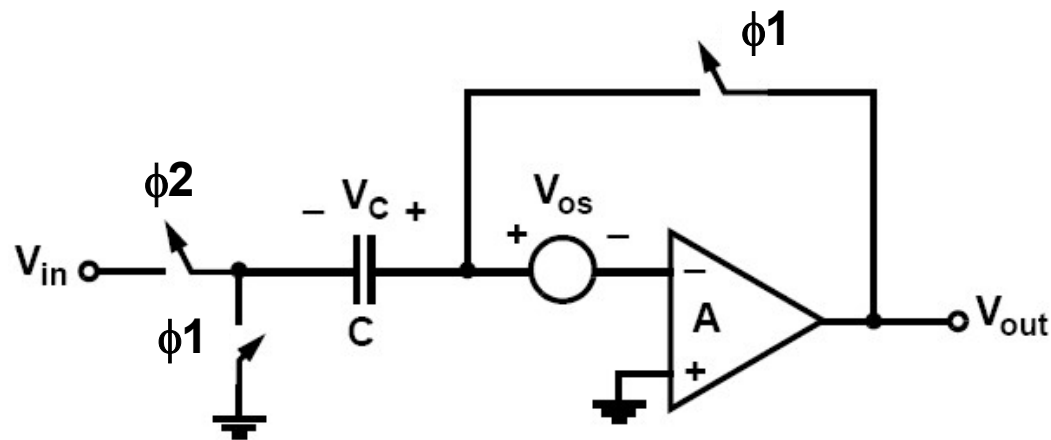
Ref: Poujois, et al., JSSC 8/78



- Phase 1: Offset storage; Phase 2: Amplify
- Design considerations
 - Must ensure that amplifier does not saturate during phase 1
 - Must make C sufficiently large to avoid attenuation and mitigate charge injection error

Input Series Cancellation

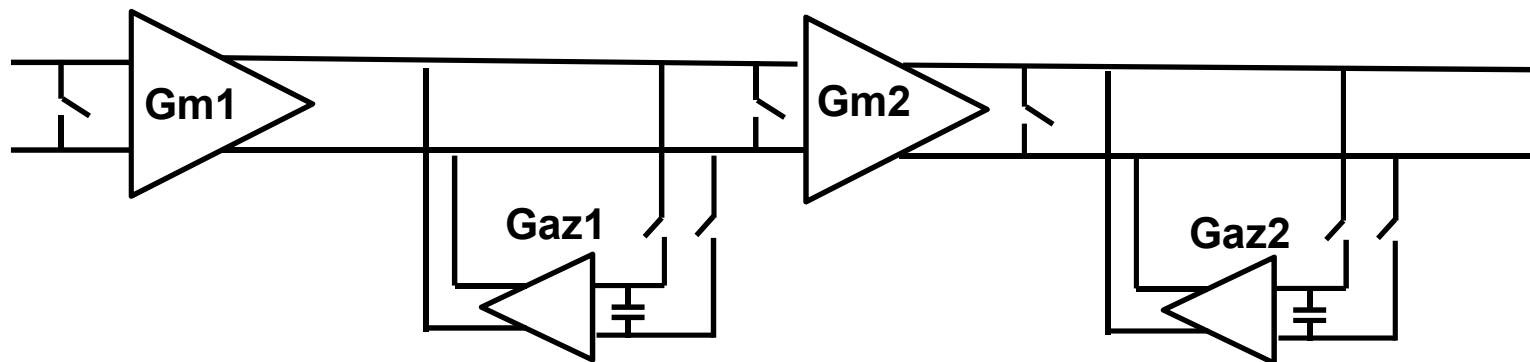
Refs: McCreary & Gray
Yee, et al.



- Phase 1: Offset storage; Phase 2: Amplify
- In phase 2, input referred offset $\approx V_{os}/(A+1)$
 - 4x reduction if $A=3$

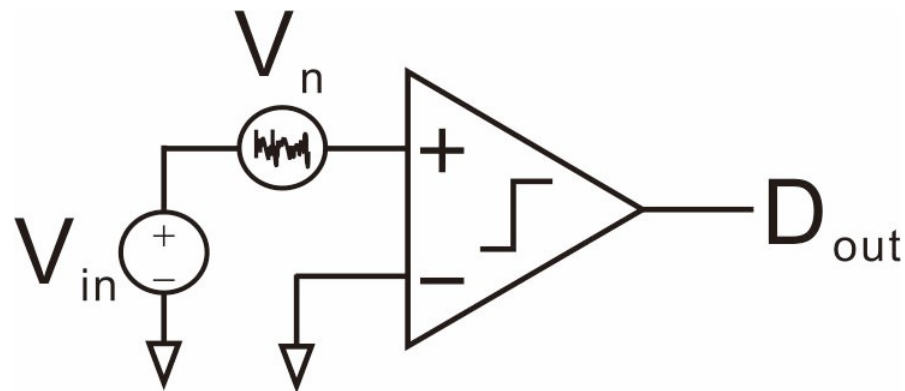
What about using an integrator as pre-amp?

- Integrator vs. pre-amp
 - Faster, higher gain, lower noise, better suppression for latch noise/offset
- Cannot use traditional input/output series offset cancellation
 - Too slow to settle due to high gain
 - Can saturate for output series offset cancellation
- Use a diode-connected gm to store the current mismatch

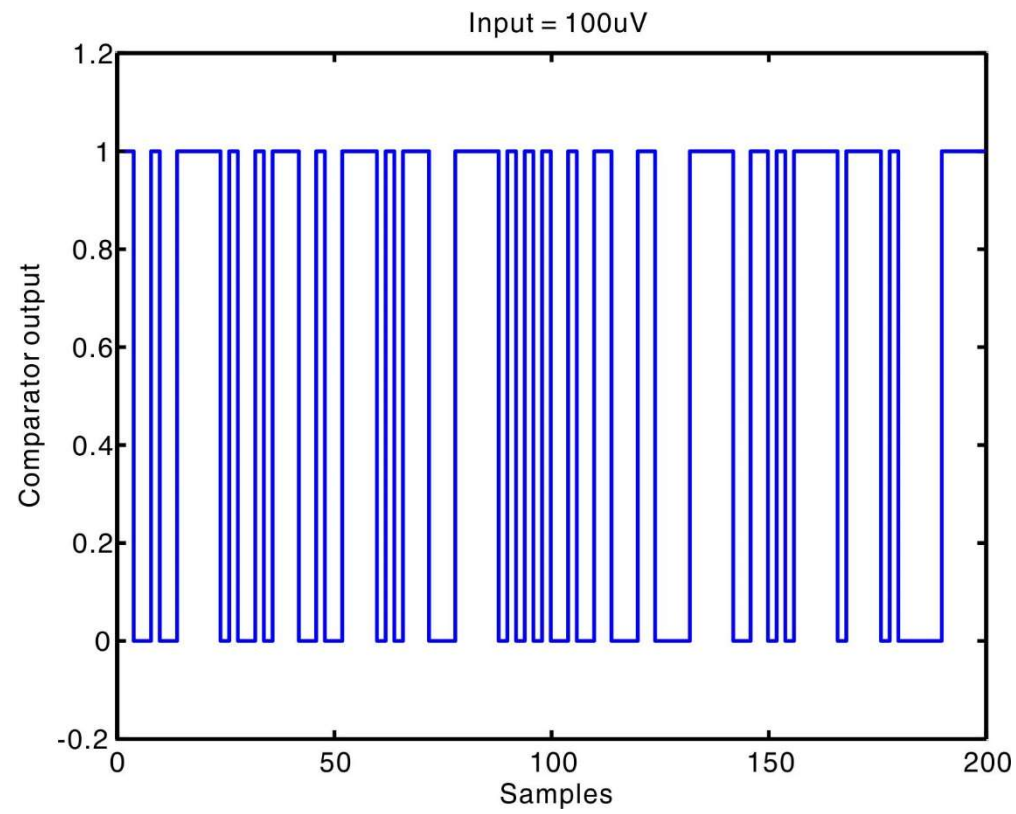


Comparator Transient Noise Simulation

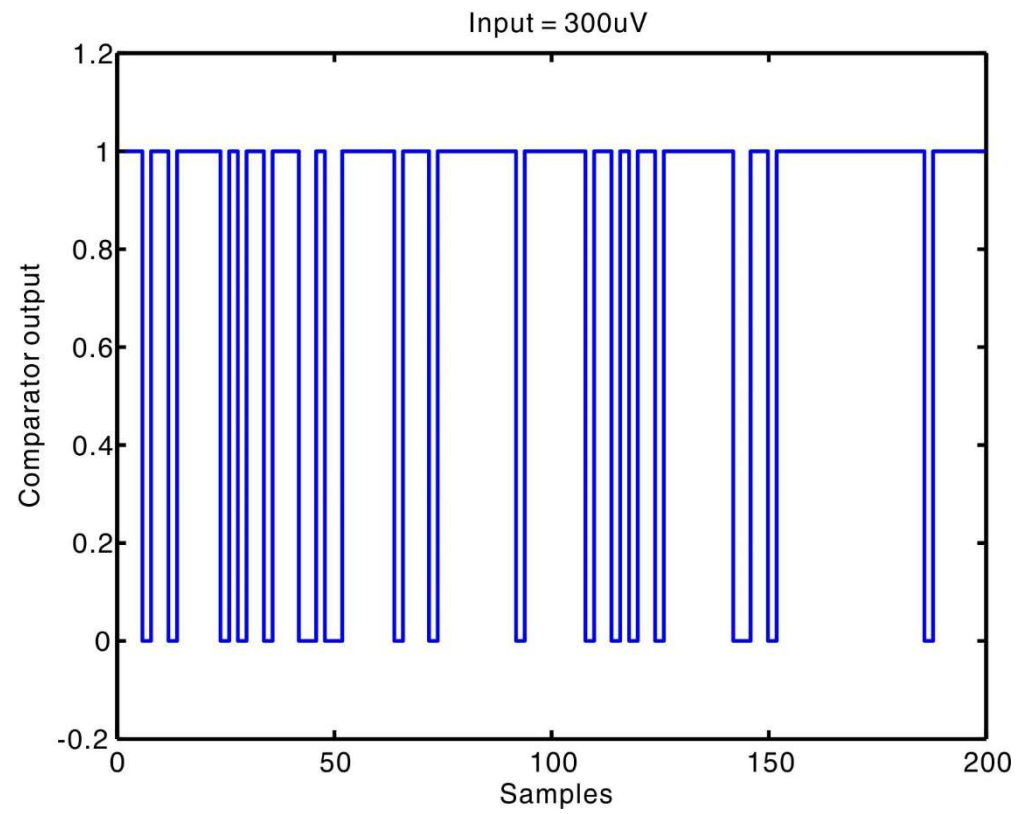
- Comparator is a dynamic circuit
 - Not LTI
 - No operating point $\rightarrow$ AC noise simulation does not apply
- Transient noise simulation works
 - We give an nonzero input, and calculate the input referred comparator noise based on the histogram



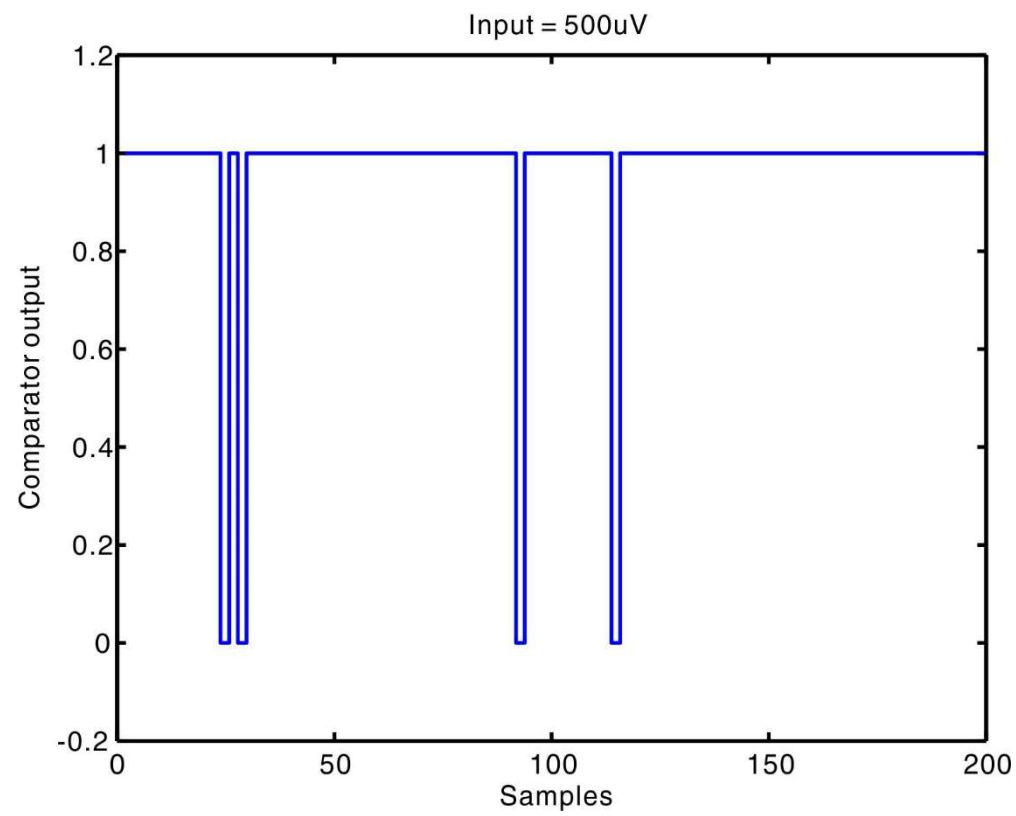
$$V_{in} = 100 \text{ } \mu\text{V}$$



$$V_{in} = 300 \text{ } \mu\text{V}$$



$$V_{in} = 500 \text{ } \mu\text{V}$$



Comparator Noise Simulation (2)

- Assuming comparator input referred noise follows normal distribution, we can estimate comparator noise σ based on the following formula:

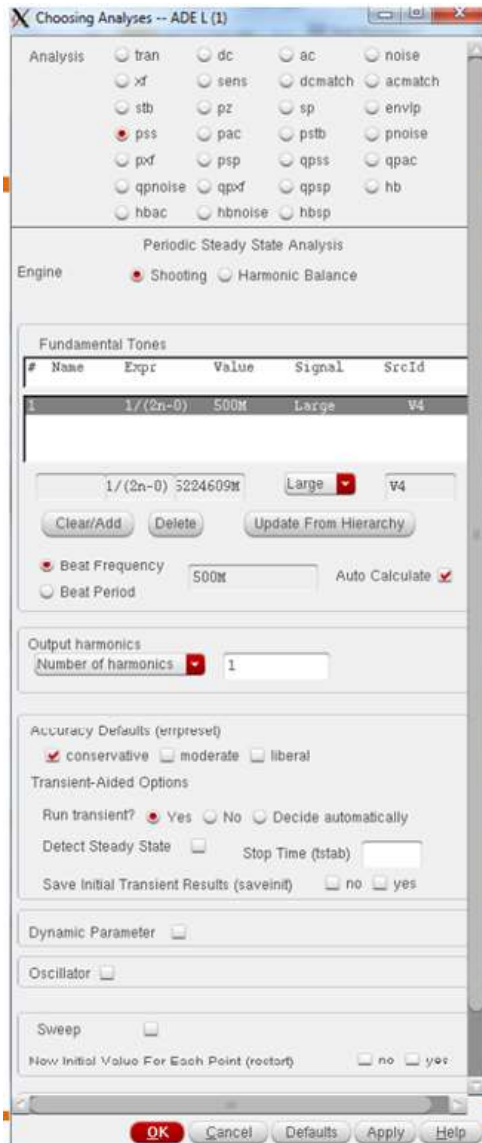
$$P(D_{out} = 1) = P(V_n < V_{in}) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{V_{in}} e^{-\frac{x^2}{2\sigma^2}} dx$$

- $V_{in} = 300\mu\text{V}$, $P(D_{out}=1) = 80\% \rightarrow \sigma \approx 350\mu\text{V}$
 - In MATLAB, $\sigma = V_{in} / \text{norminv}(P, 0, 1)$
- To reduce estimation error, we want to:
 - Simulate more samples
 - Make $V_{in} \approx \sigma$
- Be careful
 - If the comparator has offset (due to post-layout asymmetry), we need to take it away first
 - We also need to ensure the comparator is fully reset (no memory effect), otherwise there will be correlation between samples, causing miscalculations

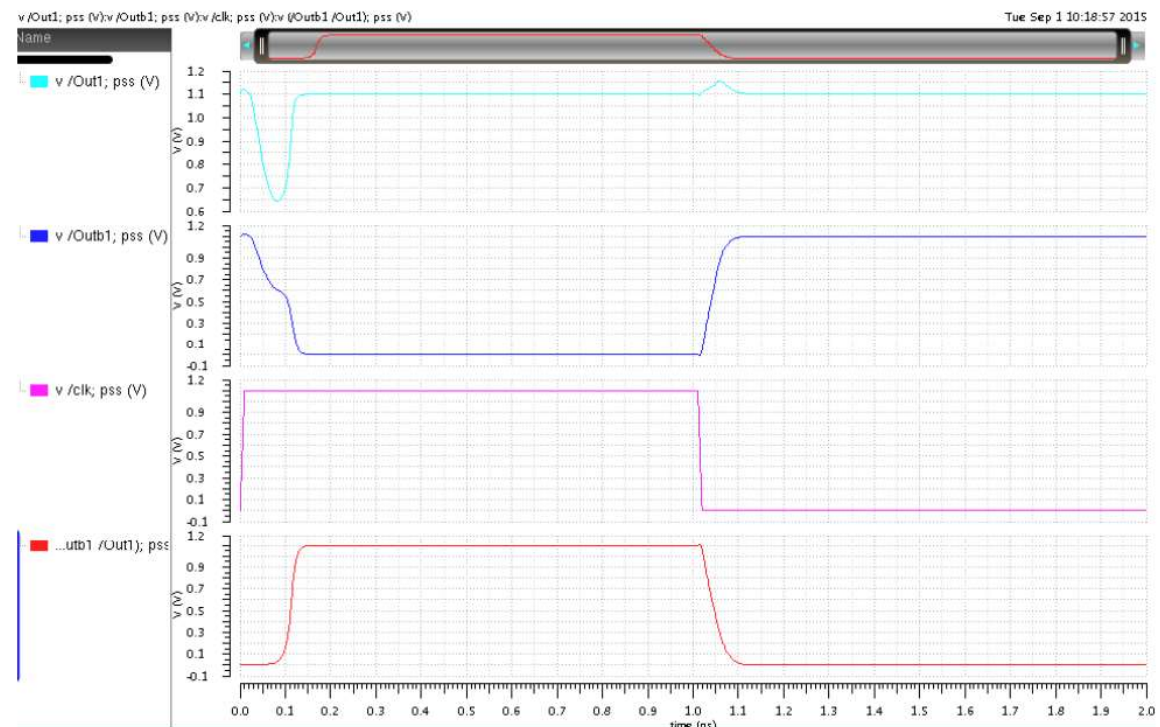
Comparator PNOISE simulation

- We can use transient noise, but it is sometimes too slow due to
 - Large noise bandwidth needed
 - In TSMC 40nm, I found that the noise bandwidth typically needs to be at least 50GHz
 - Long simulation time (at least 100+ points are needed)
- Yet, there are still randomness, which can cause the noise estimation to fluctuate
- It would be nice if we can get PSS PNOISE simulation to work
 - Although there is no steady-state for a comparator especially when its input is very small, it seems that the time-domain PNOISE simulation is correct.

PSS/PNOISE

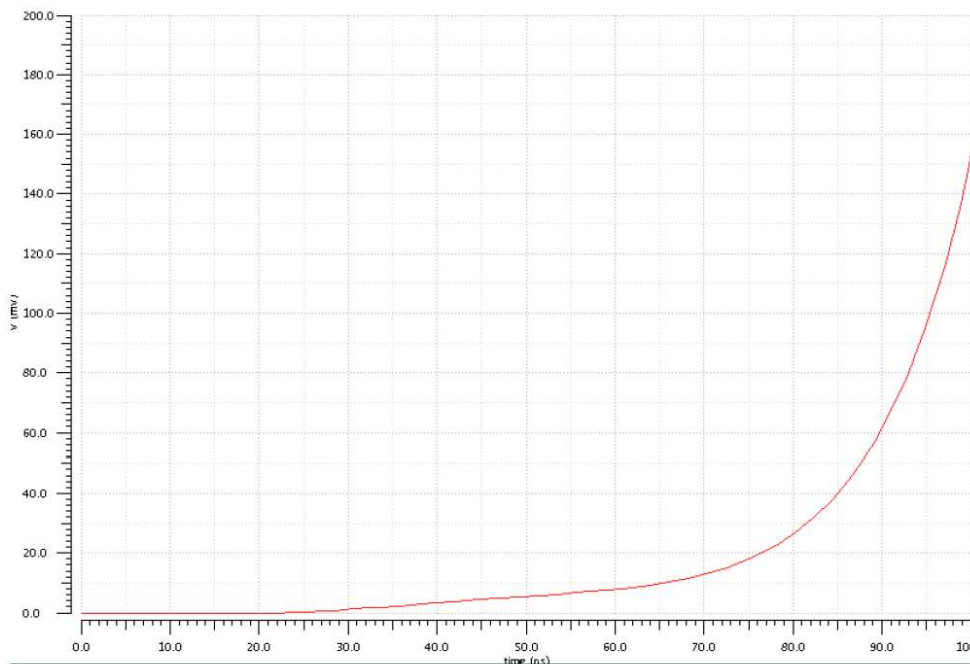


- We can set up the comparator and give a very small input (e.g., 1mV).
- Then we run PSS simulation to get steady state (without the presence of noise)



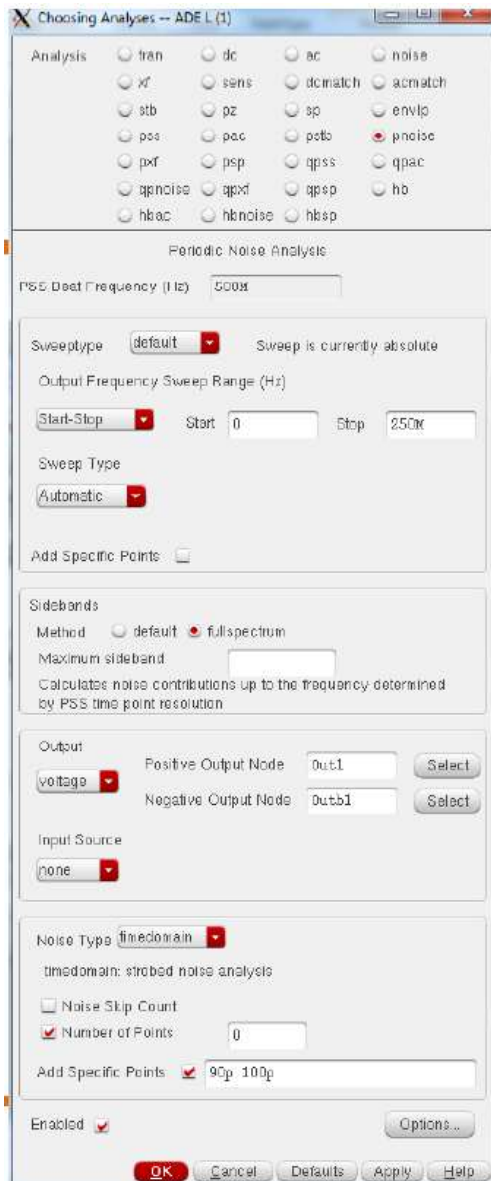
PSS/PNOISE

- We can calculate the gain of the comparator at specific times instances
 - We can also estimate comparator time constant
 - If we can calculate the total time-domain integrated noise, then we know the total input referred noise



Time	Gain
70p	13
80p	26
90p	61
100p	152

Run PNOISE



- PNOISE gives us the noise at specific time points
 - Here I am calculating the noise at 90ps and 100ps
- In sidebands, now we can choose “full spectrum”
 - It automatically calculates the number of sideband needed
 - It uses APS so the simulation time is very short

PNOISE Simulation Results

- The input referred noise does not change much with time
- As long as we are in regeneration phase, the total input referred noise remains roughly the same
- Why?
 - In regeneration phase, both signal and noise grows at the same rate
 - Can be derived using Langevin equation with negative resistance
- Runtime for 6 time points
 - Only 4 seconds

Time	Gain	Total output noise [mV]	Total input referred [mV]
50p	5.4	7.7	1.4
60p	7.8	9.7	1.2
70p	13	15	1.1
80p	26	29	1.1
90p	61	66	1.1
100p	152	161	1.1

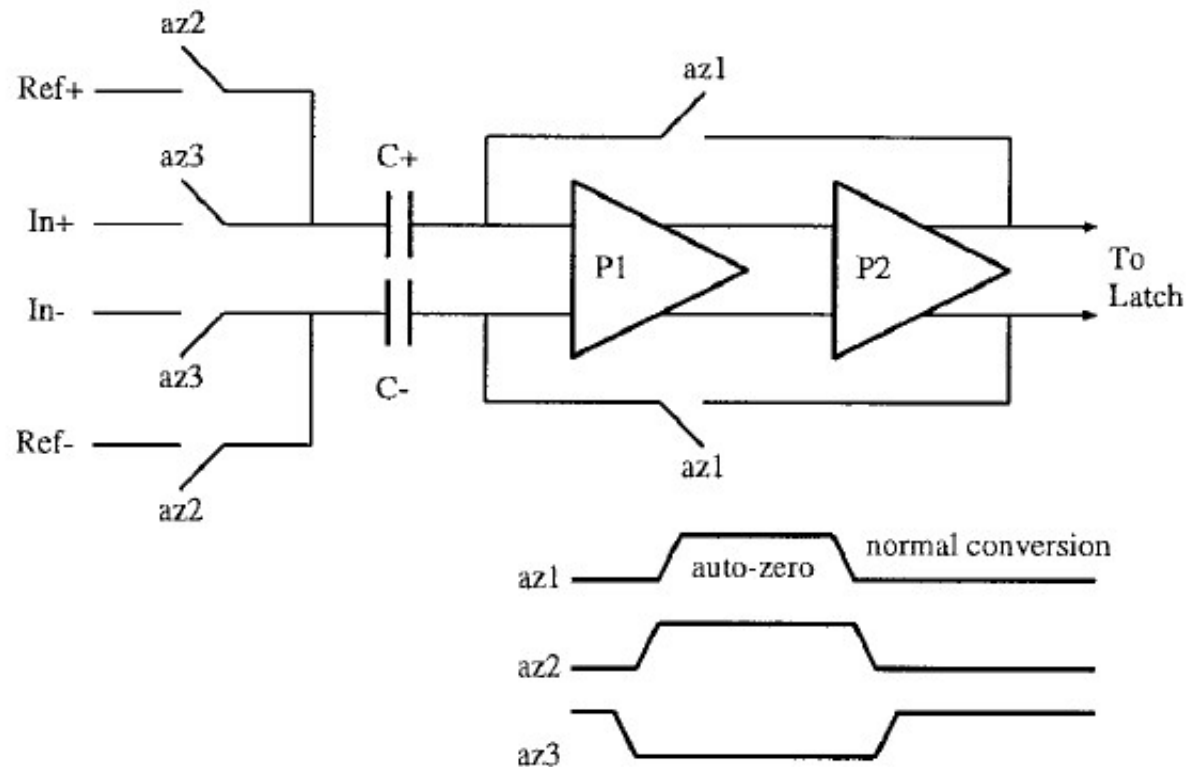
Transient noise simulation

Noise BW	Run Time [s]	Flip-rate	Input referred [mV]
10G	18	0.12	0.85
20G	23	0.12	0.85
40G	32	0.15	0.97
80G	49	0.15	0.97

- Using APS for speed-up
 - Without APS, 40G BW would take 84s, flip rate 0.15
- Simulating 500 points
 - There is still fluctuations in the output, and we still underestimate the noise
- In comparison, PSS/PNOISE is 10x faster
 - Can be used in comparator synthesis

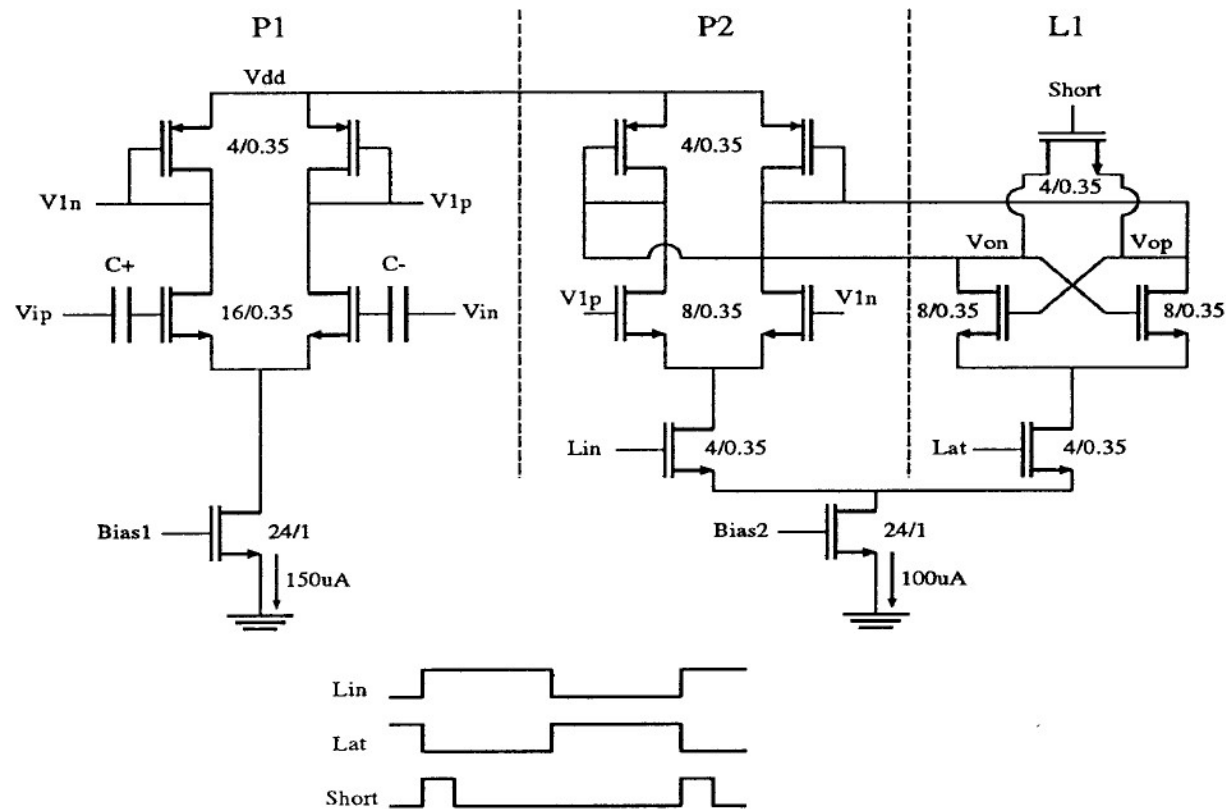
Static Comparator #1

- Mehr & Dalton, JSSC 7/1999



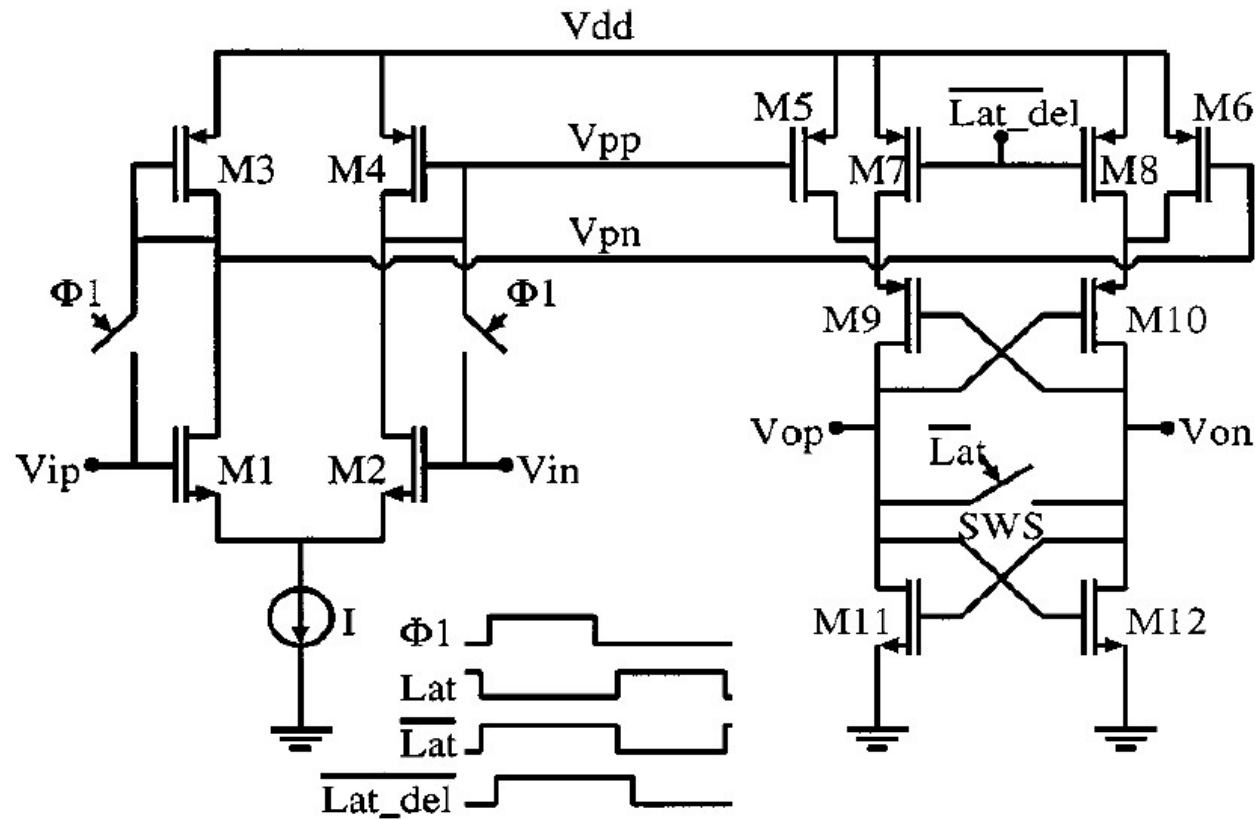
Static Comparator #1

- Mehr & Dalton, JSSC 7/1999



- Two stage gain; current re-use; short the output to clear charge

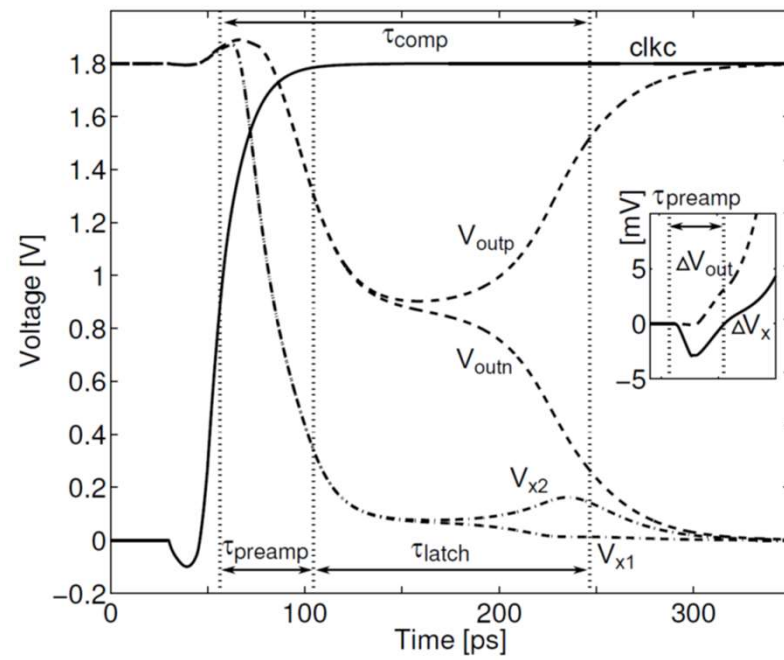
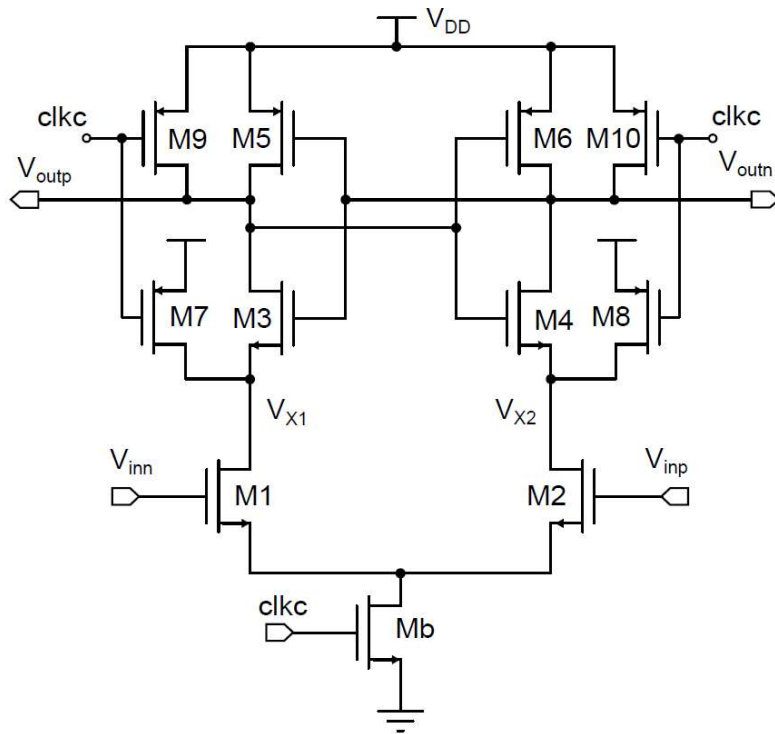
- Mehr & Singer, JSSC 3/2000



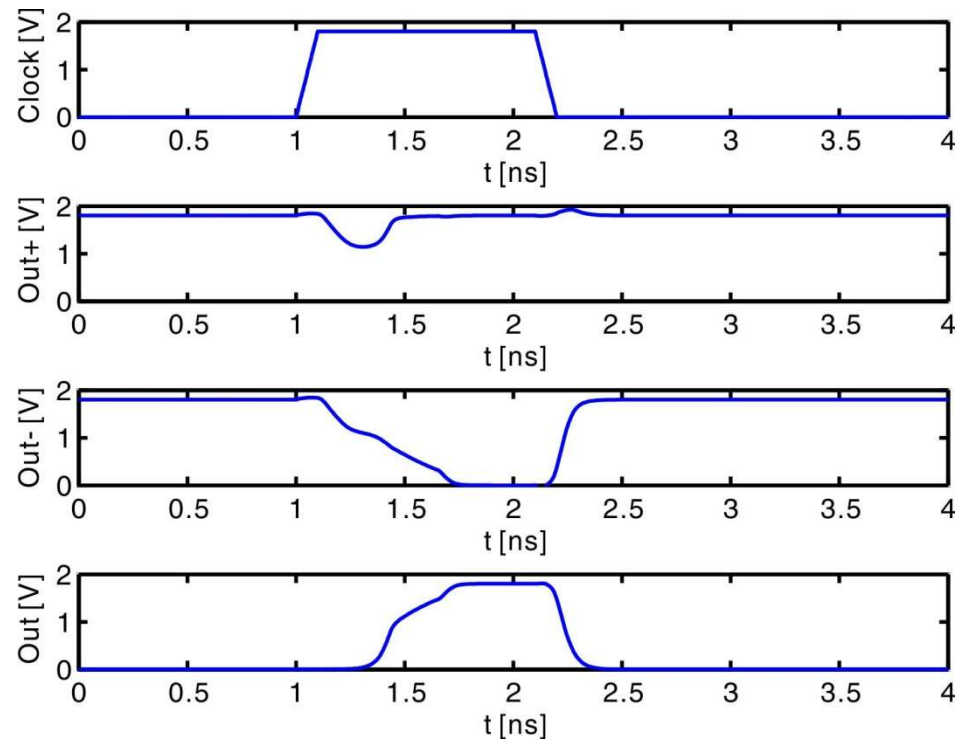
- Input series cancellation; provide input common mode; couple the current

Strong-Arm Latch

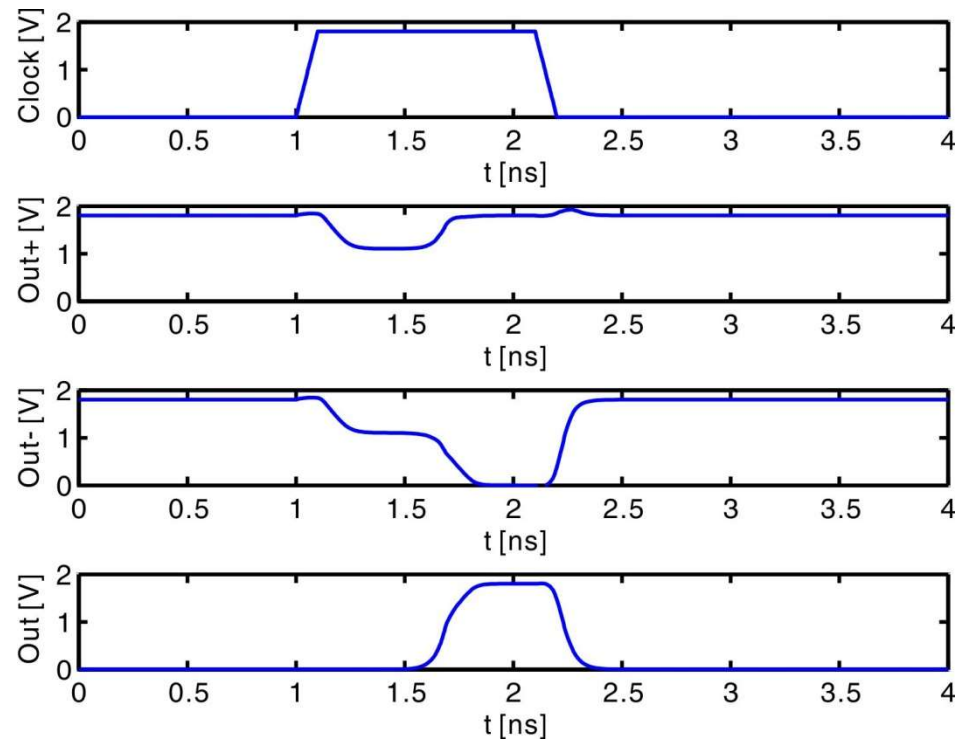
Fully-dynamic: no static current



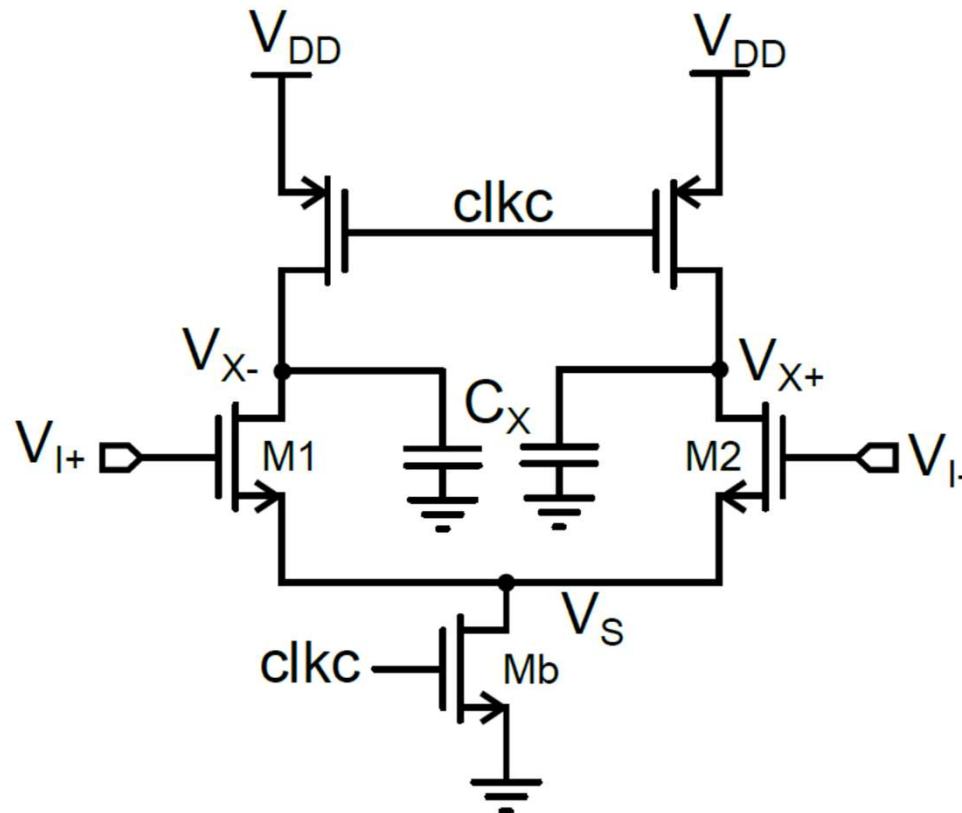
Simulation for $V_{in} = 1\text{mV}$



Simulation for $V_{in} = 1\mu V$

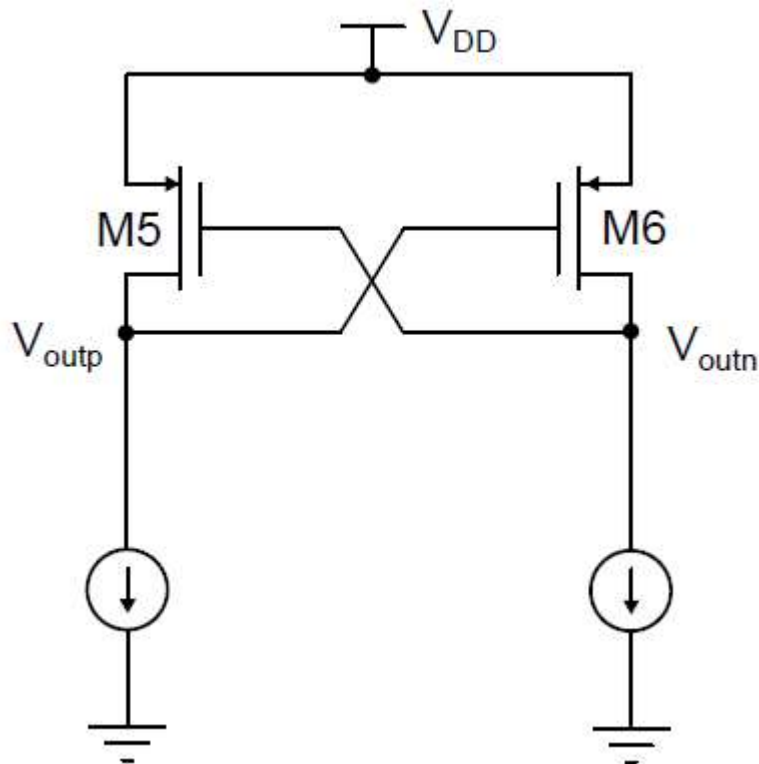


Pre-amplification phase: dynamic integrator



- $G = \frac{g_m}{I_D} \cdot V_{tn}$
- $\overline{\sigma_{n,preamp}^2} = \frac{4kT\gamma}{g_m/I_D} \cdot \frac{1}{C_x V_{tn}}$

Latch phase



- Both signal and noise grows exponentially
- Latch input referred noise does not decrease with time

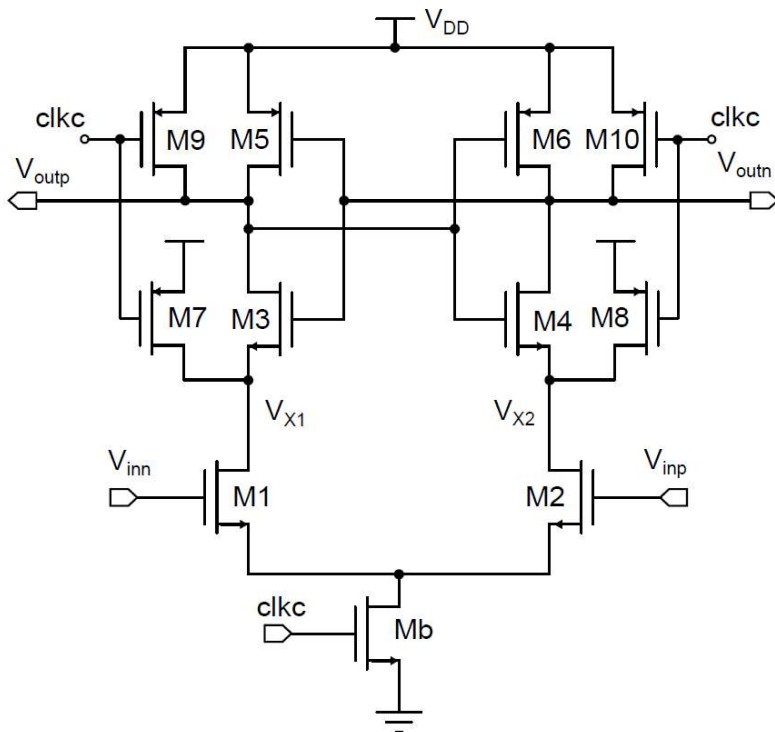
$$\overline{\sigma_{n,latch}^2} = \frac{2(1 + \gamma)kT}{C_o}$$

Comparator noise and offset analysis

$$\sigma_n = \sqrt{\sigma_{n,preamp}^2 + \frac{\sigma_{n,latch}^2}{G^2}} \quad \sigma_{os} = \sqrt{\sigma_{os,preamp}^2 + \frac{\sigma_{os,latch}^2}{G^2}}$$

- For a well-optimized comparator, we prefer:
 - Large G : attenuate the noise from the latch stage
 - For Strong-Arm, G is 3~5
 - Large g_m/I_D : it improves energy efficiency
 - In practice, g_m/I_D is 10~20 V^{-1}
 - A too-large g_m/I_D slows down the comparator and increases the comparator input capacitance

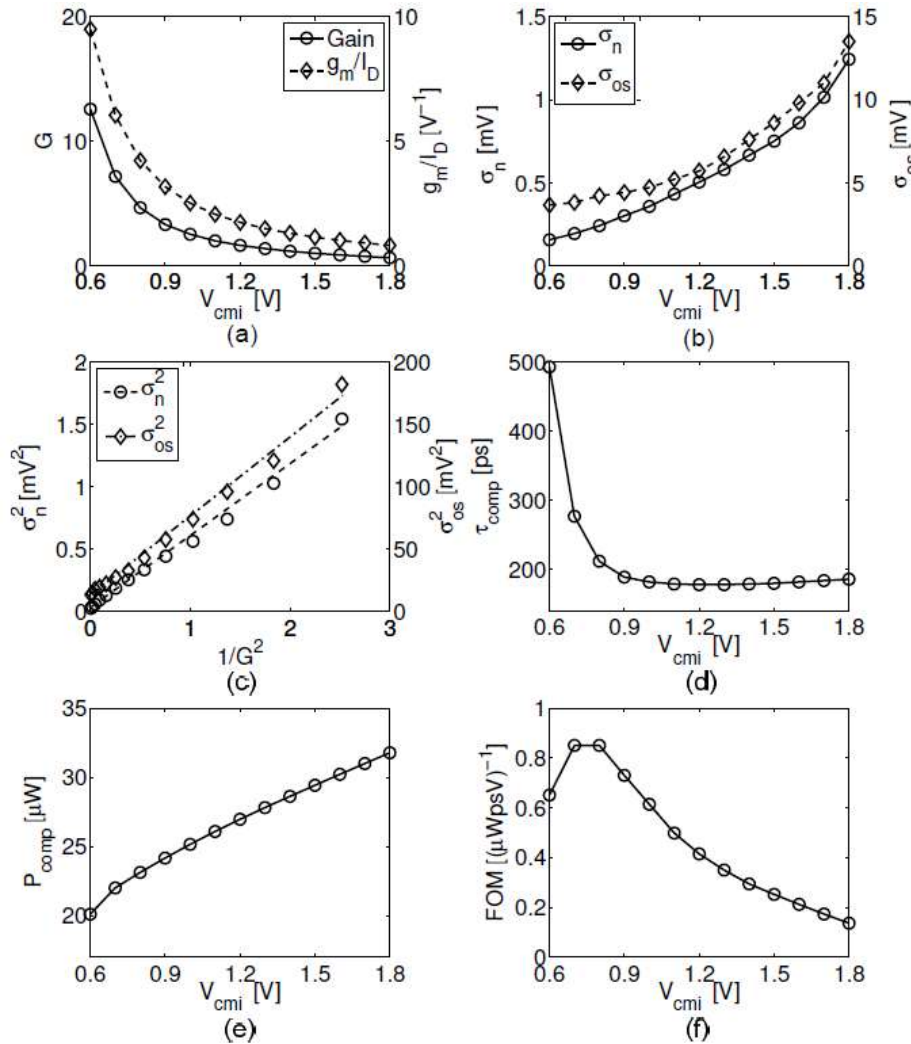
Strong-Arm Latch



- Dynamic pre-amp
- No static current
- Tail transistor operates in linear region
- Common-mode variation
- Process corner variation
- Speed issue with low V_{cmi}
- Large room for energy efficiency boost

Impact of V_{cmi} and Process Corner

- A smaller V_{ov} increases g_m/I_D and G , but it slows down the comparator
- Comparator offset and noise increases as G decreases



Two-stage comparator v1

- Decouple regeneration from pre-amp; increase speed ☺
- M10/11 starts from linear region → low preamp gain ☹

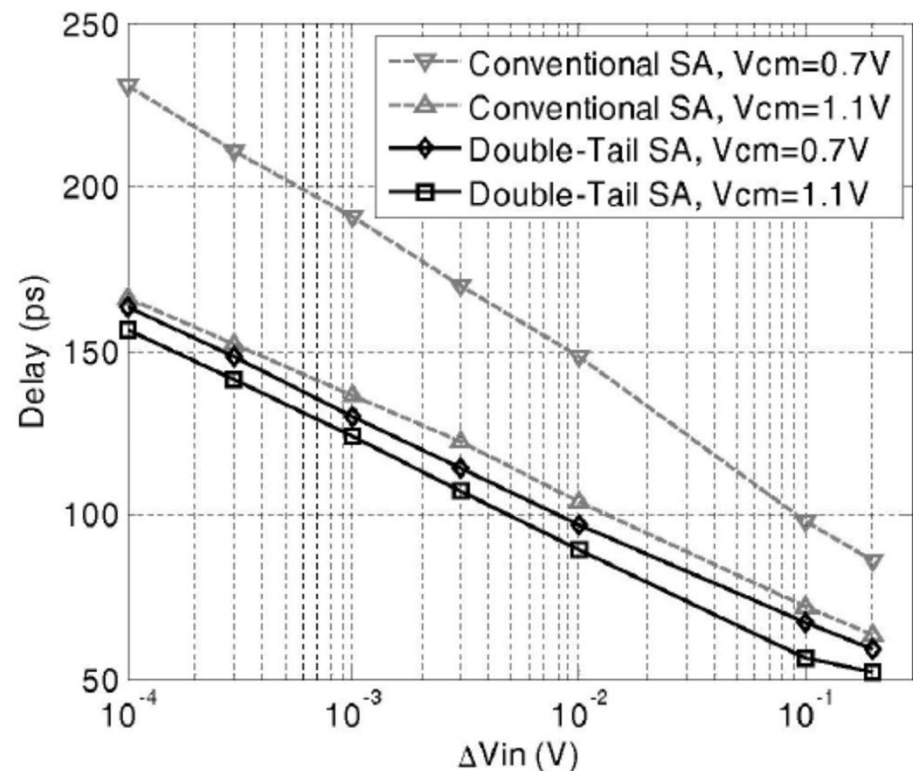
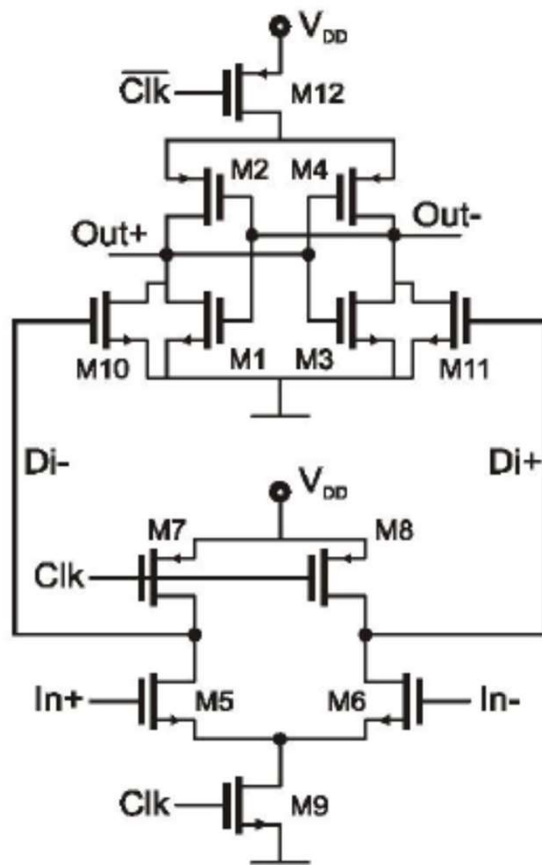
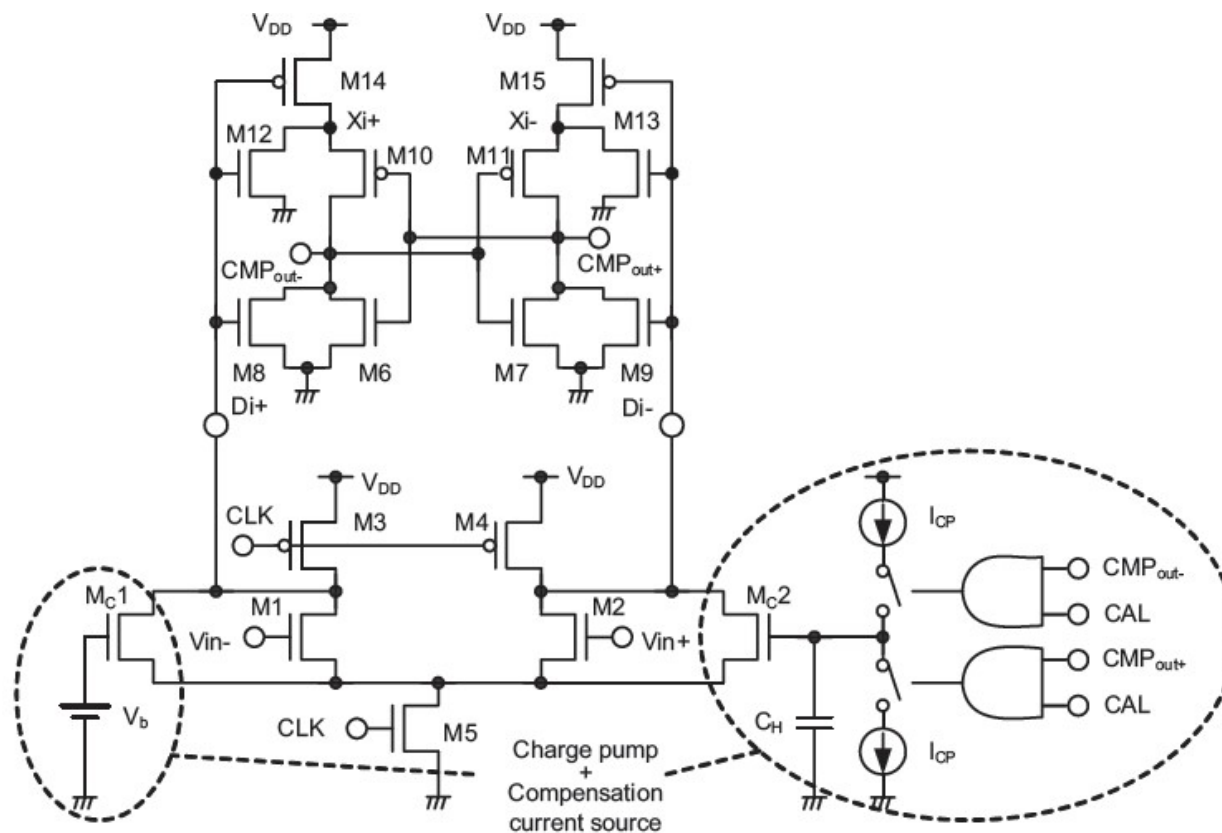


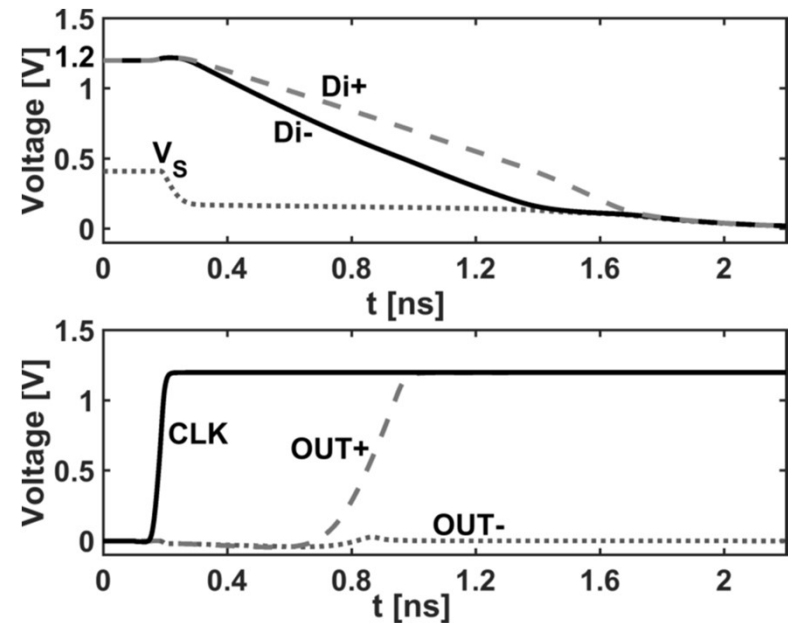
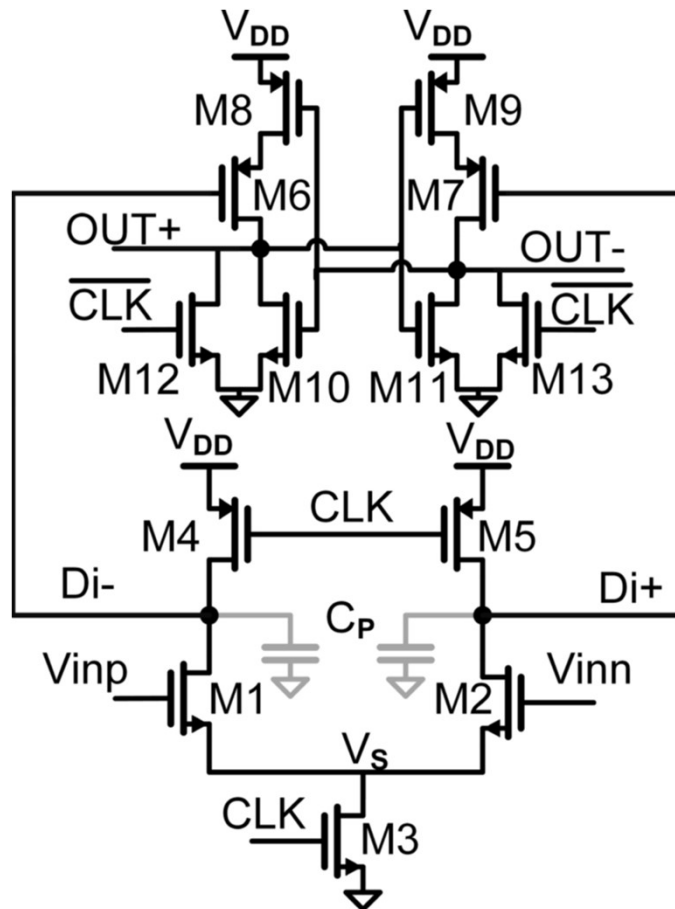
Figure 17.7.3: Simulated sense amplifier delays versus differential input voltage. The delay is the time between the clock edge and the instant when ΔOut crosses $1/2 V_{DD}$.

Two-stage comparator v2

- M14/15 provide extra gain 😊
- M14/15 starts from saturation → increased preamp gain 😊

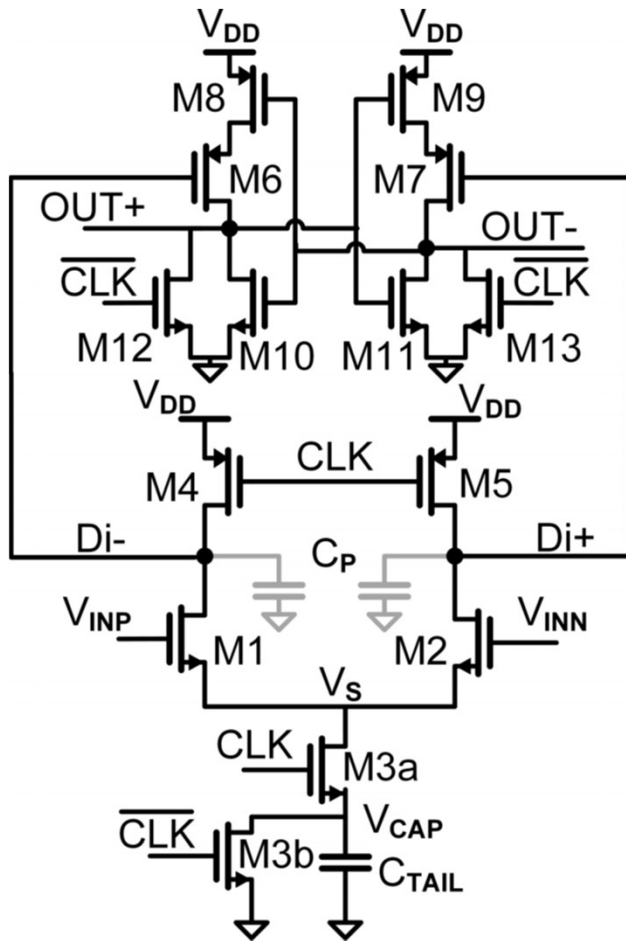


Two-stage comparator v3

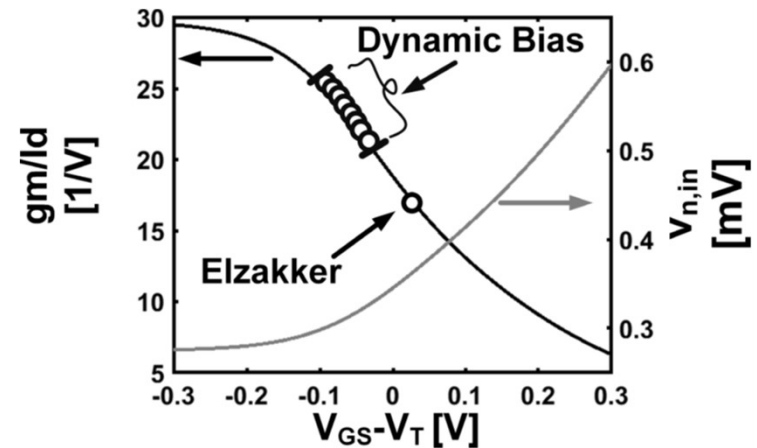
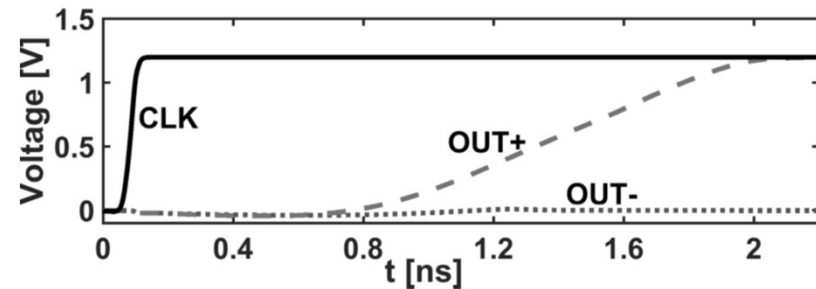
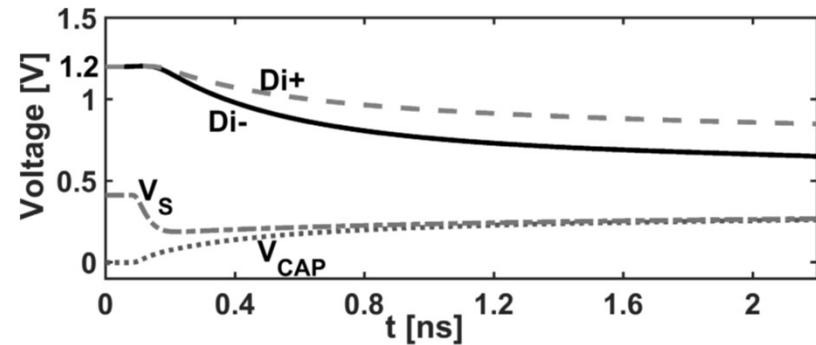


- M6/7 starts from saturation → increased preamp gain 😊
- C_p fully discharged → waste of energy ☹️

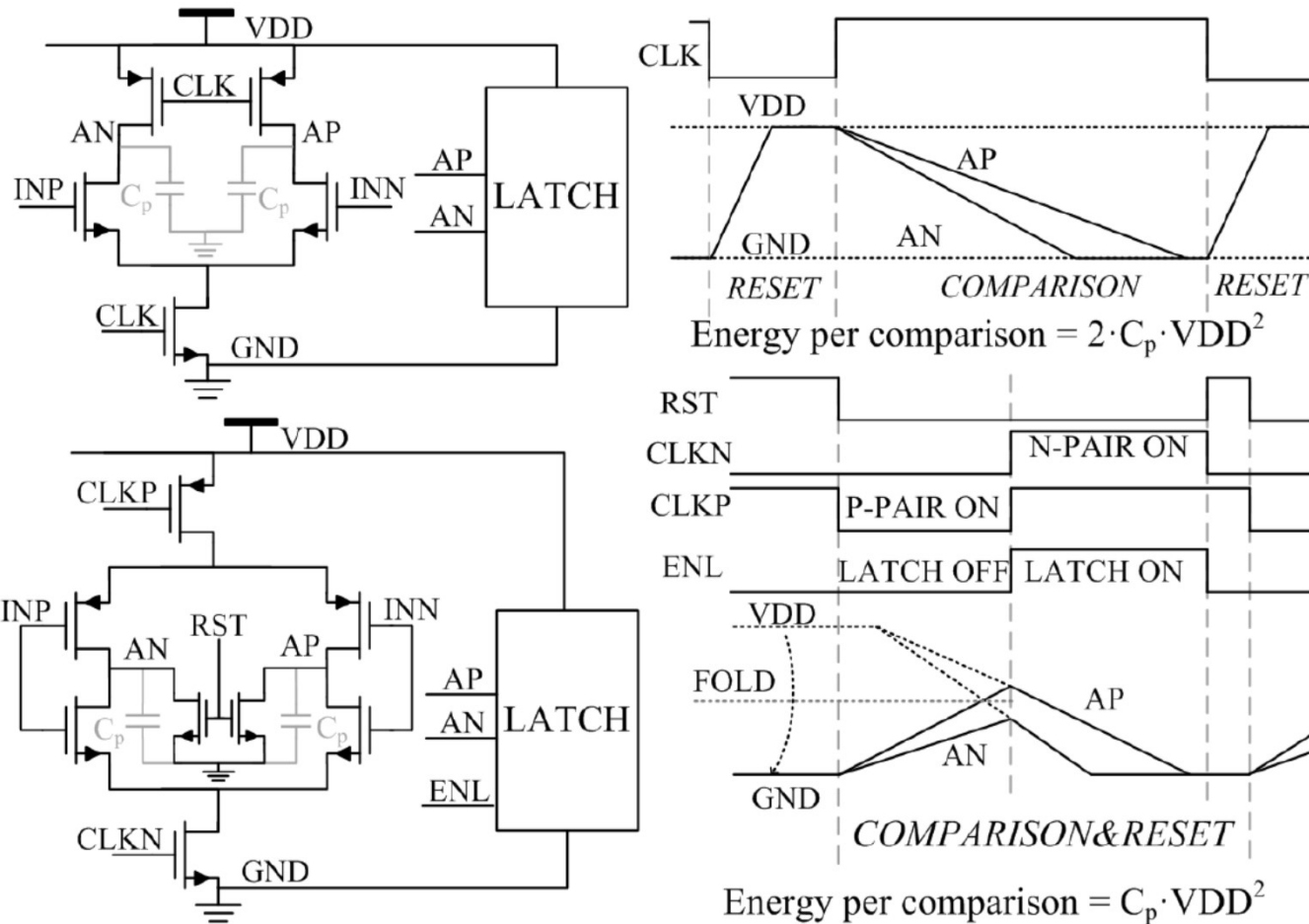
Dynamic biasing



- C_p only partially discharged ☺
- g_m/I_D increases as time goes by ☺



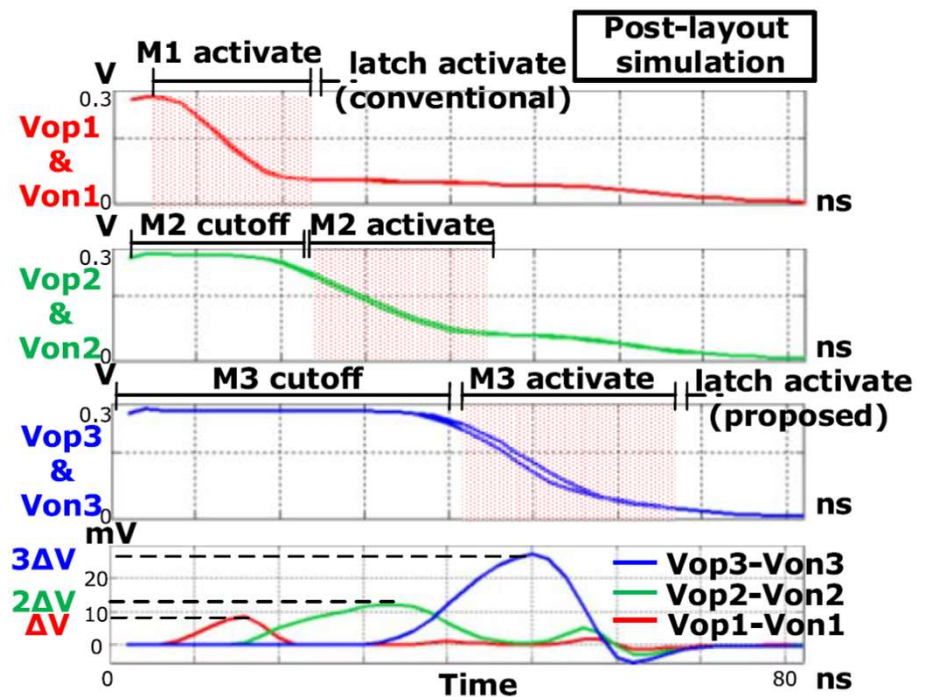
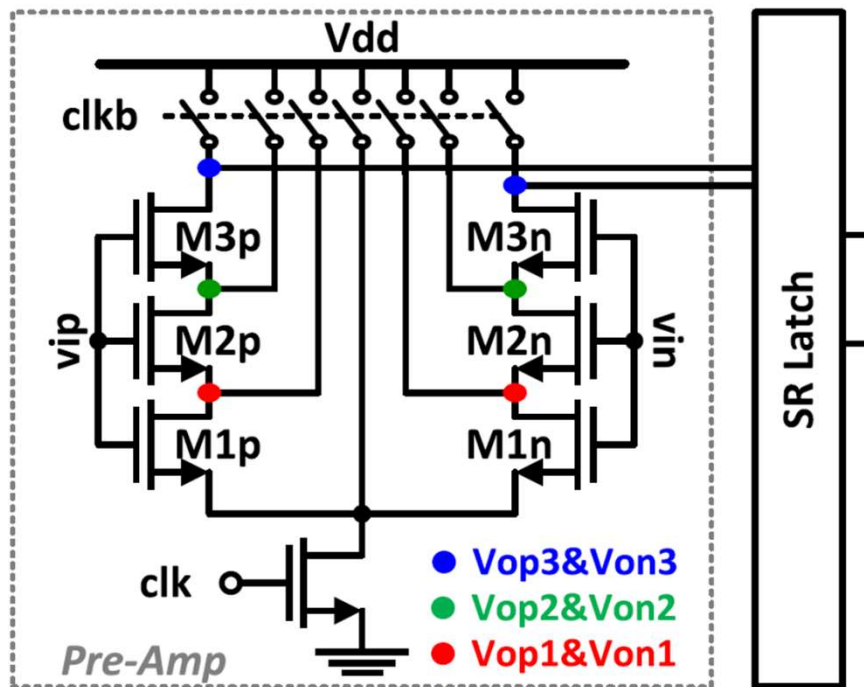
Bidirectional integration



- 2 times current reuse 😊
- C_p fully discharged ☹️

limited pre-amp gain ☹️₆₁

Cascade-input comparator



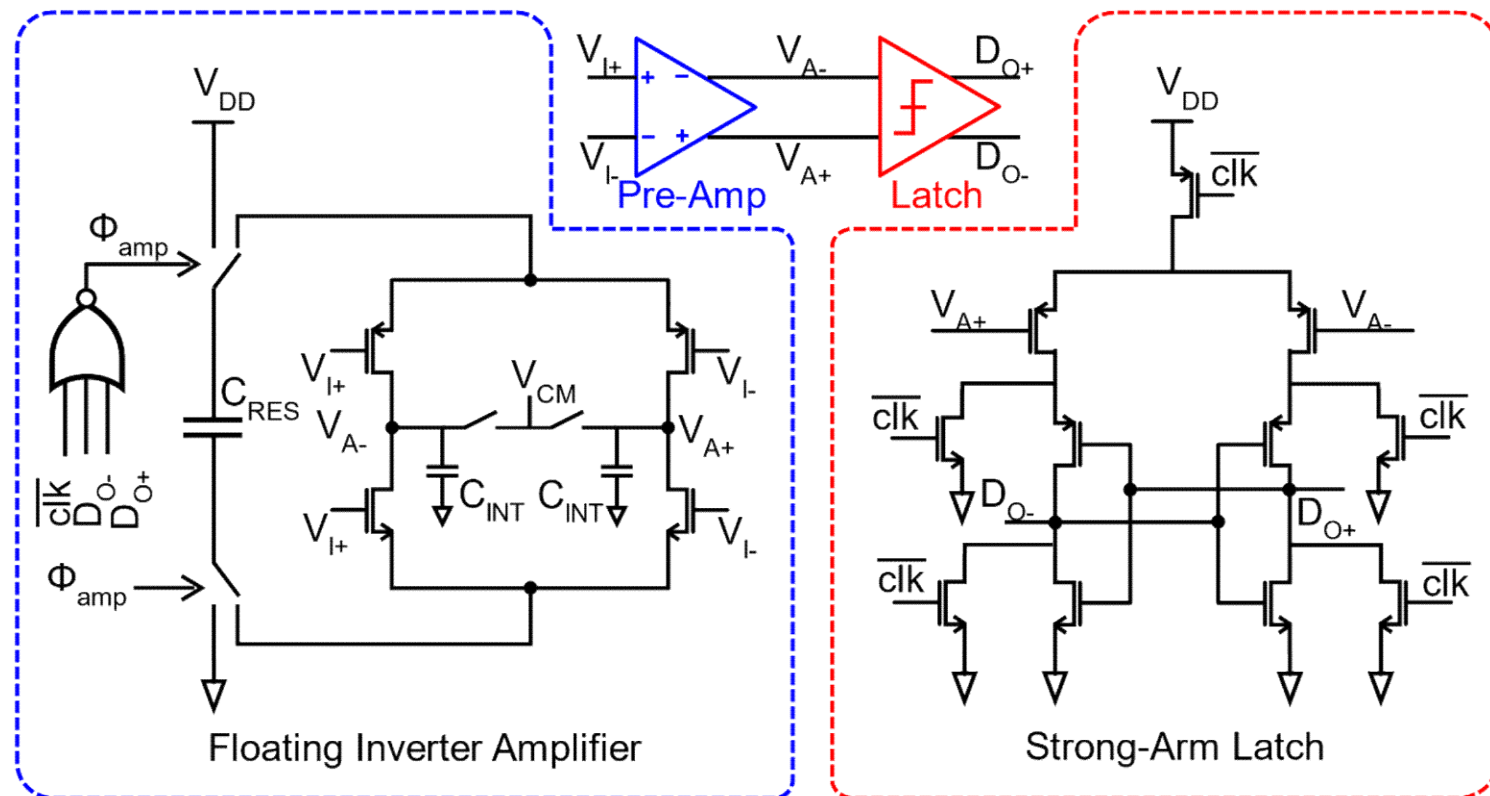
- Triple integration → 3X preamp gain 😊
- C_p fully discharged ☹️
- No net energy efficiency boost for the preamp ☹️

Objective

	Energy-efficiency				Robustness
	Current reuse	Partially discharge	Dynamic bias	Pre-amp gain	CM/Process insensitivity
Strong-Arm	☹	☹	☹	☹	☹
2-stage v1	☹	☹	☹	☹☹	☹
2-stage v2	☹	☹	☹	☺	☹
2-stage v3	☹	☹	☹	☺	☹
Dynamic Bias Int.	☹	☺	☺	☺	☹
Bidirectional Int.	☺	☹	☹	☹	☹
Cascade Input	☹	☹	☹	☺☺	☹
Objective	☺	☺	☺	☺☺☺	☺

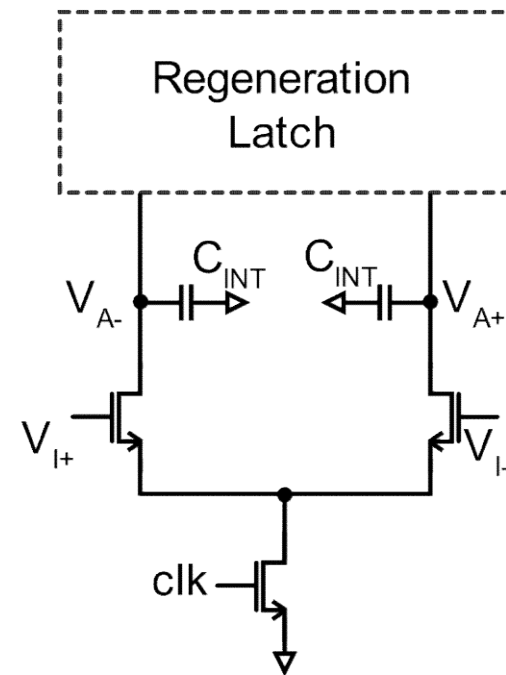
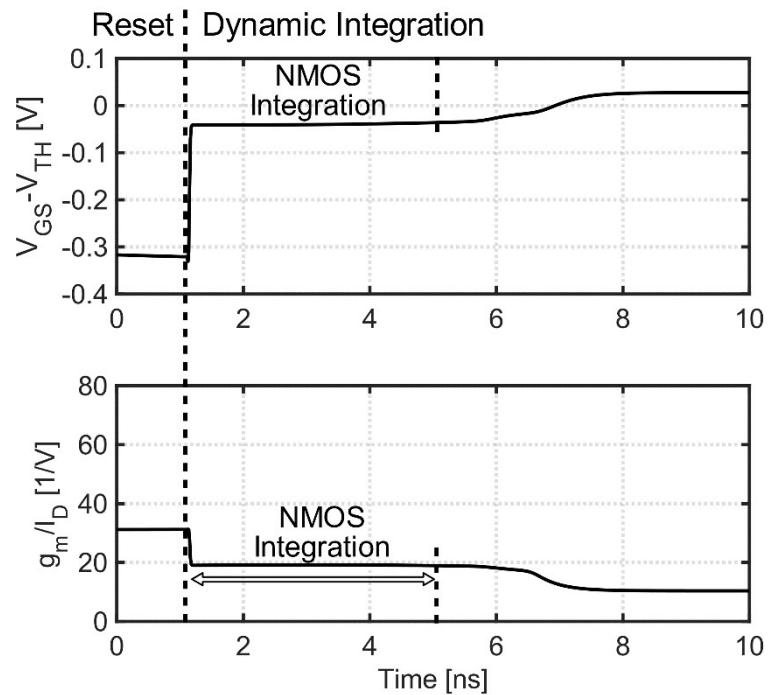
Schematic

- Floating Inverter Pre-Amplifier (FIA) + Strong-Arm Latch



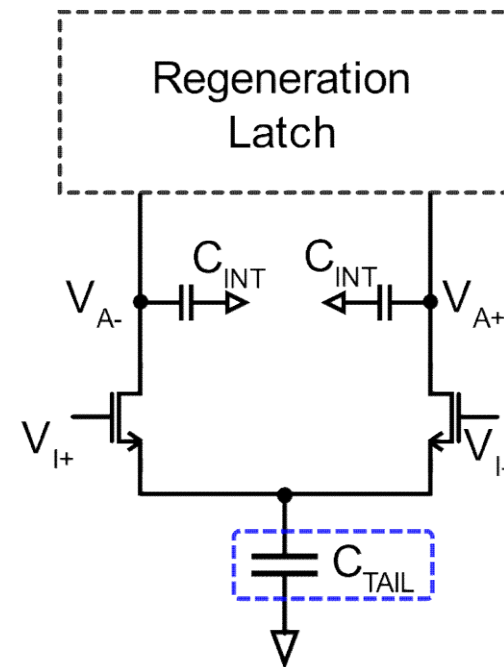
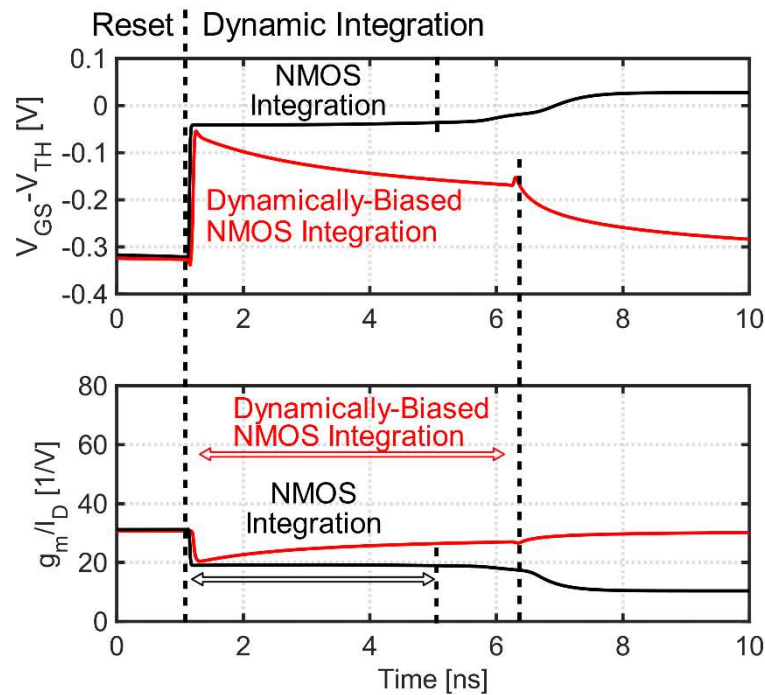
Conventional NMOS Integration

- Conventional NMOS Integration



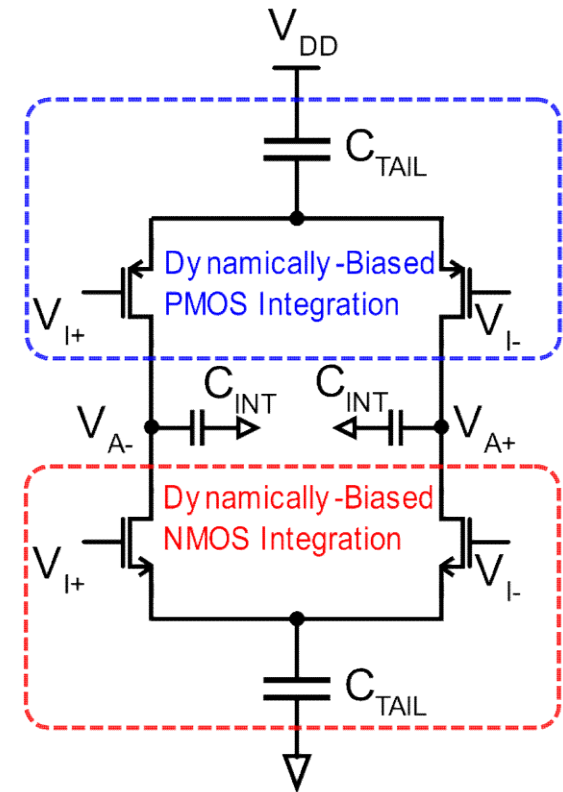
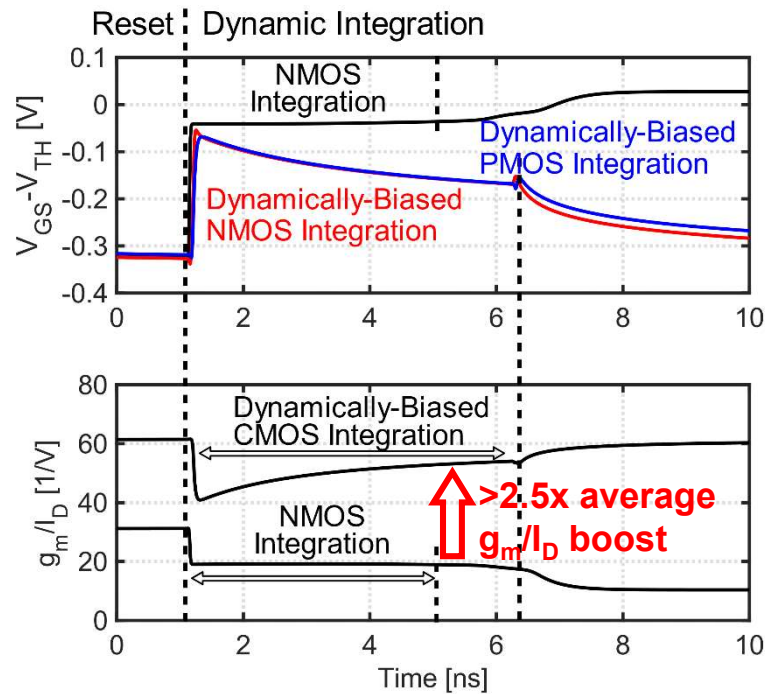
Dynamically-biased NMOS Integration

- Dynamically-biased NMOS integration
 - C_{TAIL} as degeneration capacitor



Dynamically-biased CMOS Integration

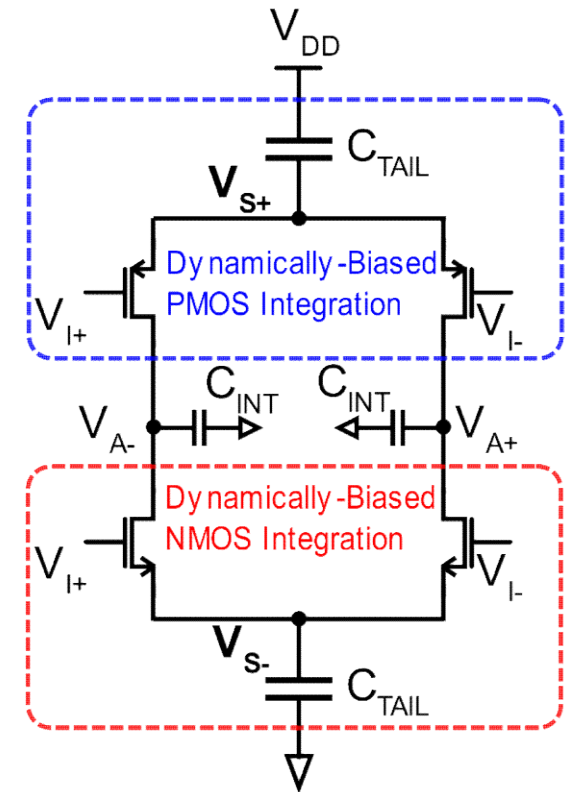
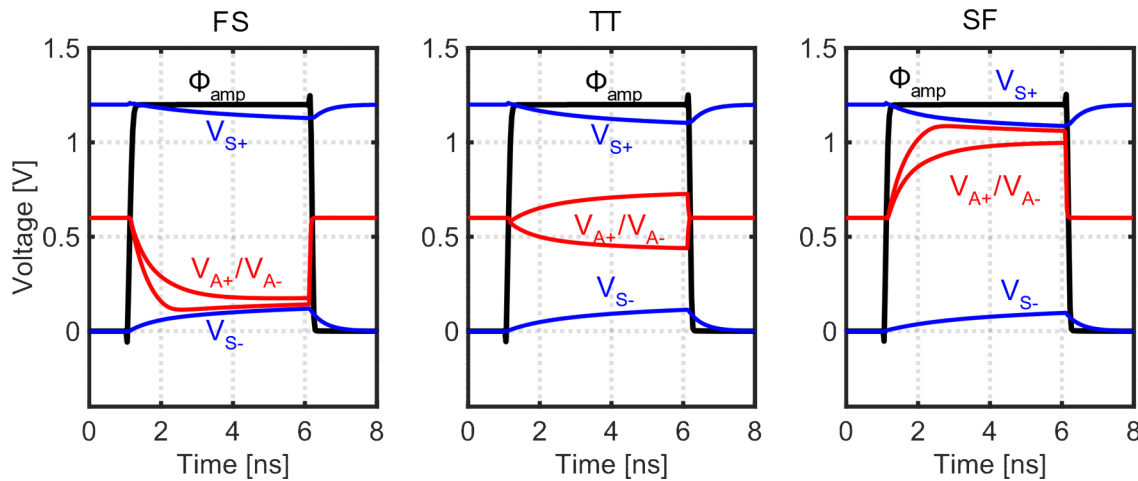
- Dynamically-biased CMOS integration
 - Current reuse 😊



Dynamically-biased CMOS Integration

- Dynamically-biased CMOS integration

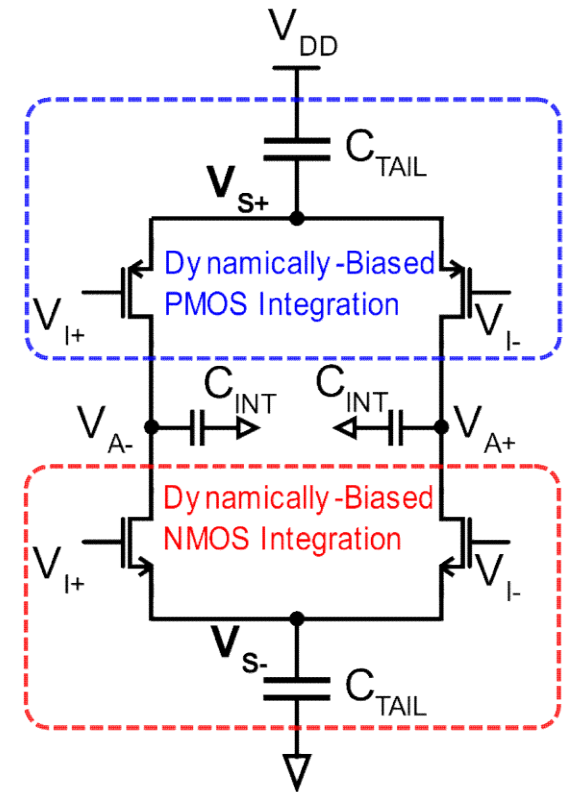
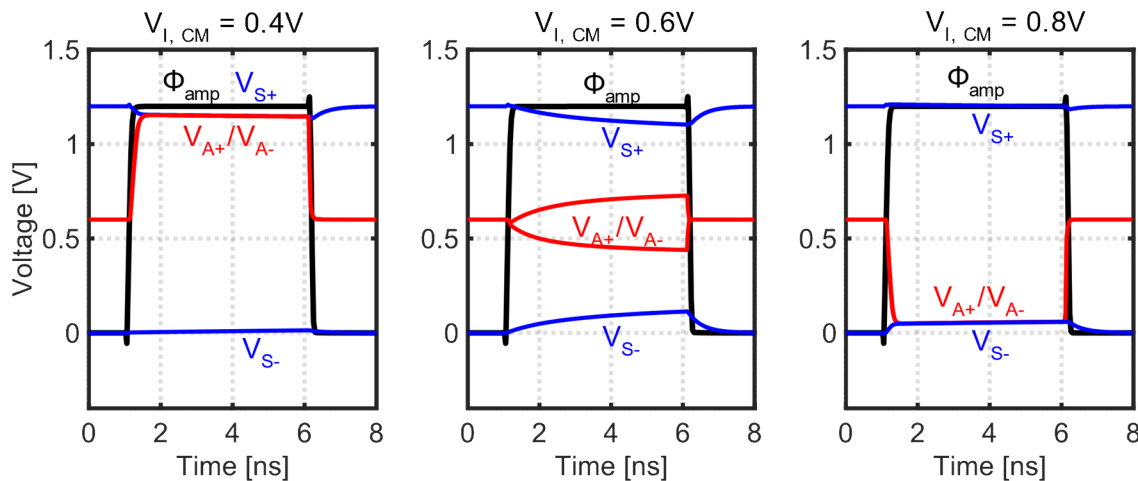
- Current reuse 😊
- Process robustness ☹️



Dynamically-biased CMOS Integration

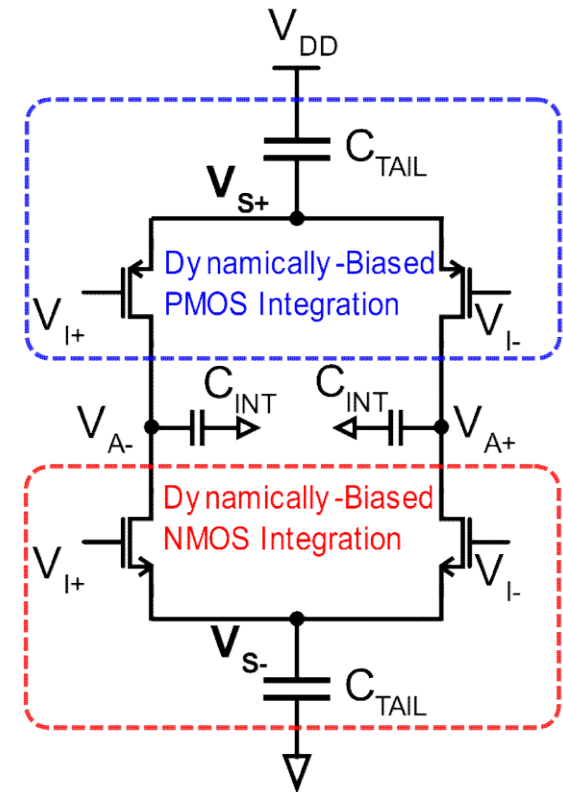
- Dynamically-biased CMOS integration

- Current reuse 😊
- Process robustness ☹️
- Input CM insensitivity ☹️



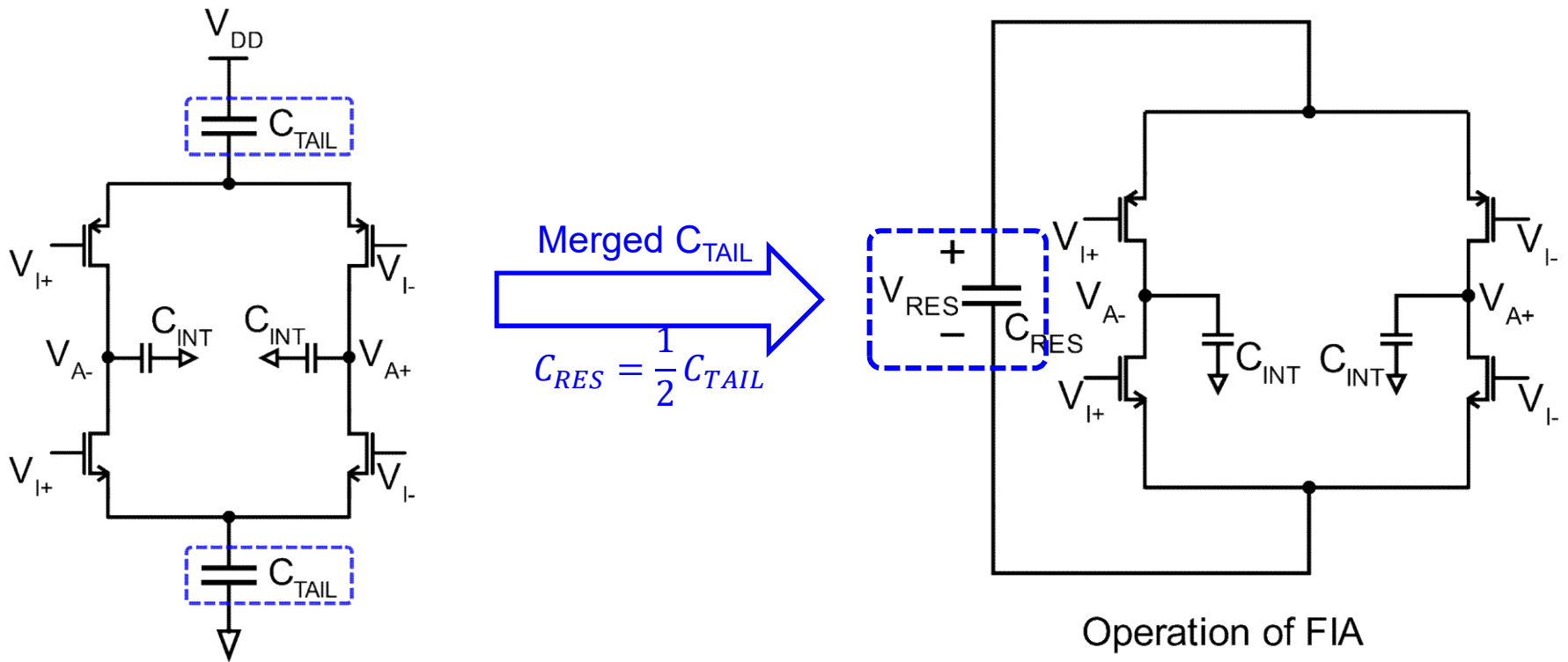
Dynamically-biased CMOS Integration

- Dynamically-biased CMOS integration
 - Current reuse 😊
 - Process robustness ☹️
 - Input CM insensitivity ☹️



Robust CMOS Integration

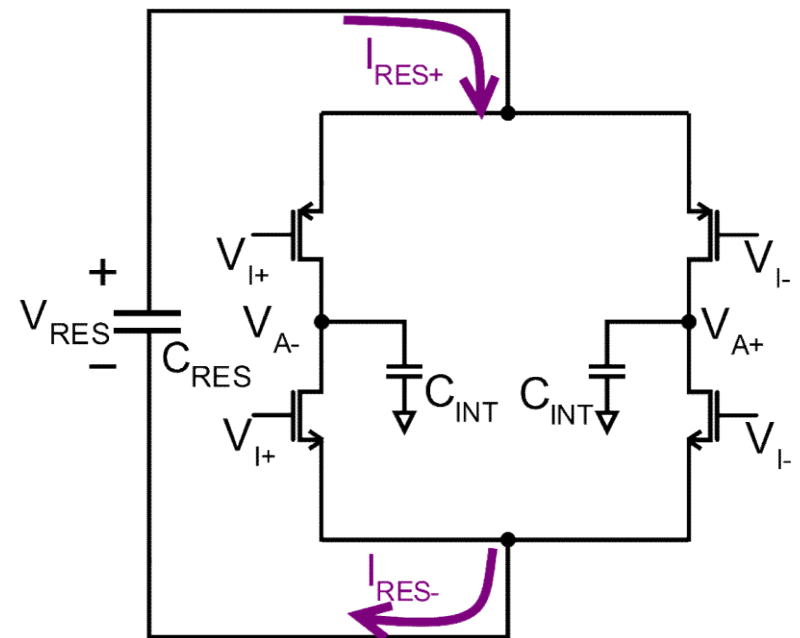
- Isolated voltage domain for FIA



Robust CMOS Integration

- Isolated voltage domain for FIA

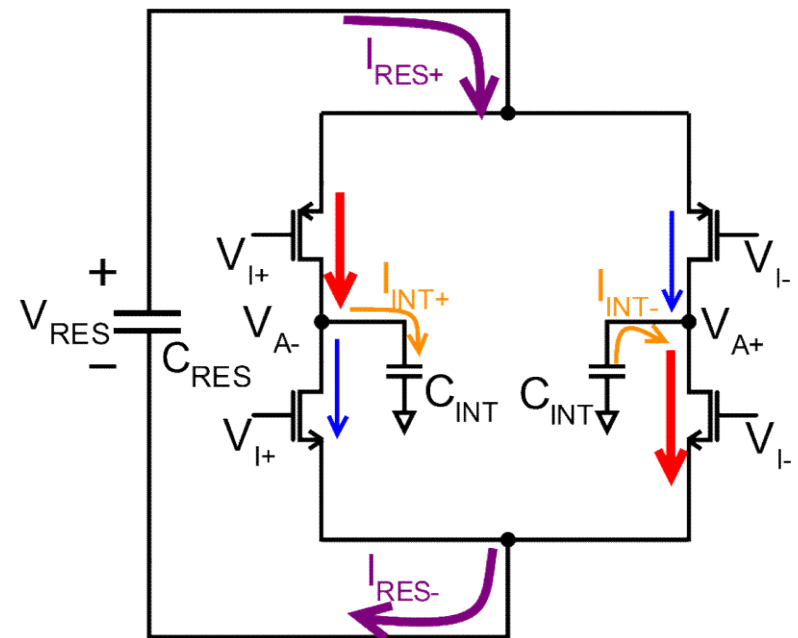
$$- I_{\text{RES}+} = I_{\text{RES}-}$$



Operation of FIA

Robust CMOS Integration

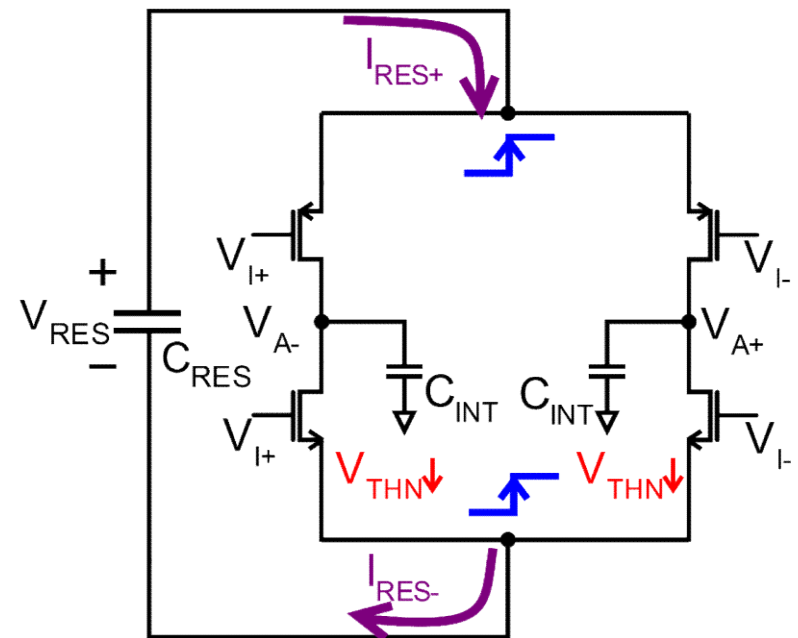
- Isolated voltage domain for FIA
 - $I_{RES+} = I_{RES-}$
 - $I_{INT+} = I_{INT-}$
 - Constant output CM voltage



Operation of FIA

Process Corner Insensitivity

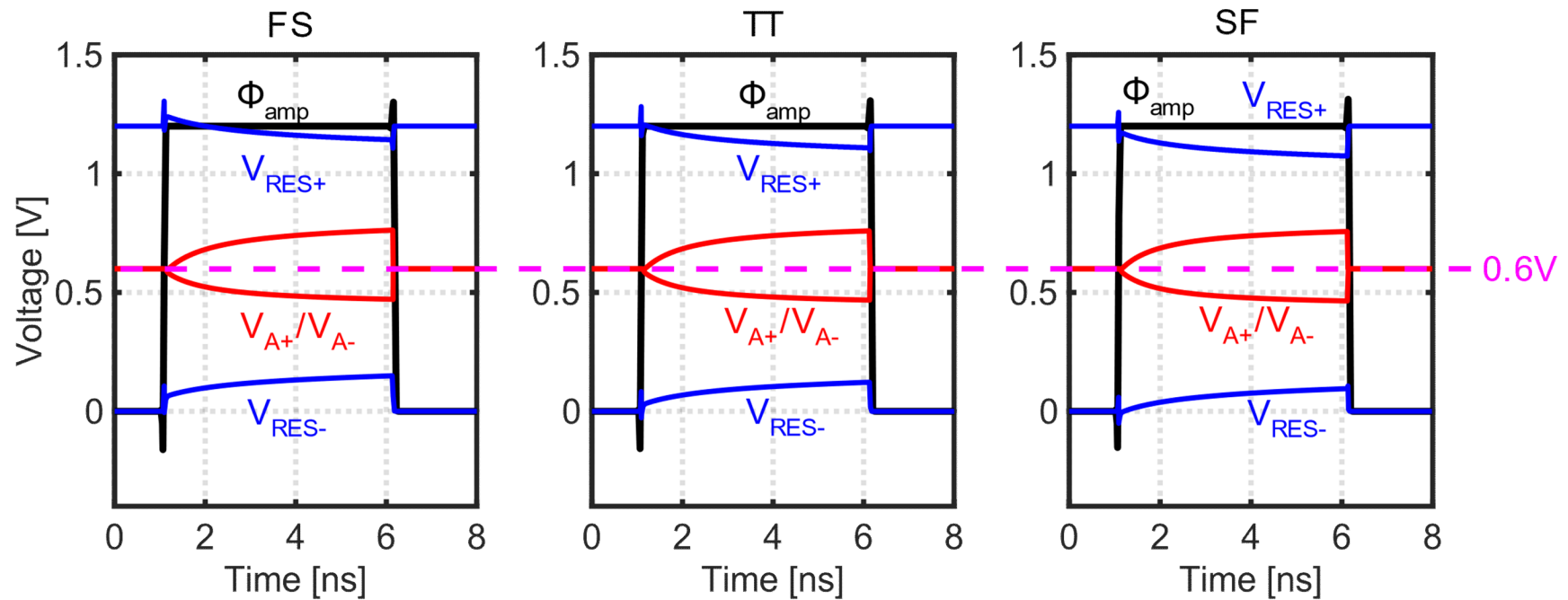
- Isolated voltage domain for FIA
 - $I_{RES+} = I_{RES-}$
 - $I_{INT+} = I_{INT-}$
 - Constant output CM voltage
- **Process corner variation**
 - Auto-balanced PMOS/NMOS strength
 - Current reuse preserved



Operation of FIA

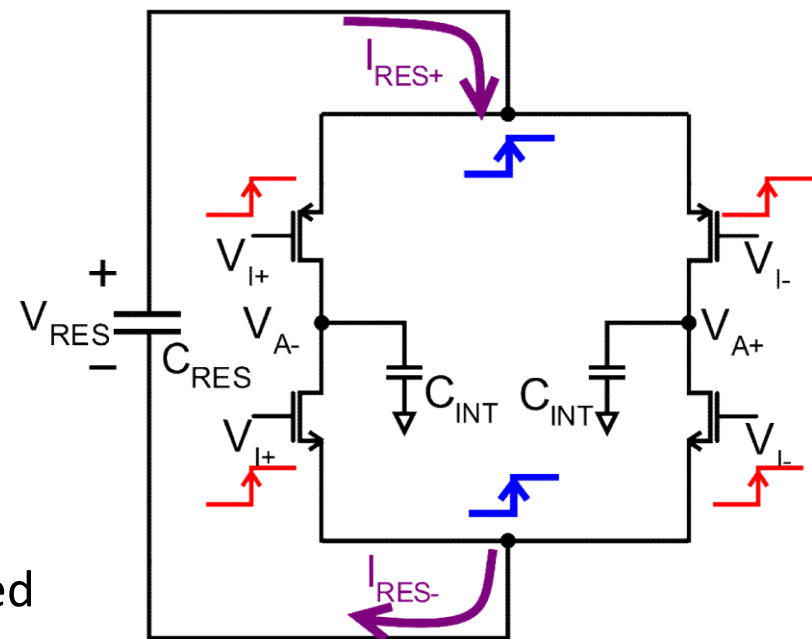
Process Corner Insensitivity

- Simulated FIA behavior with process corners



Input Common-Mode Insensitivity

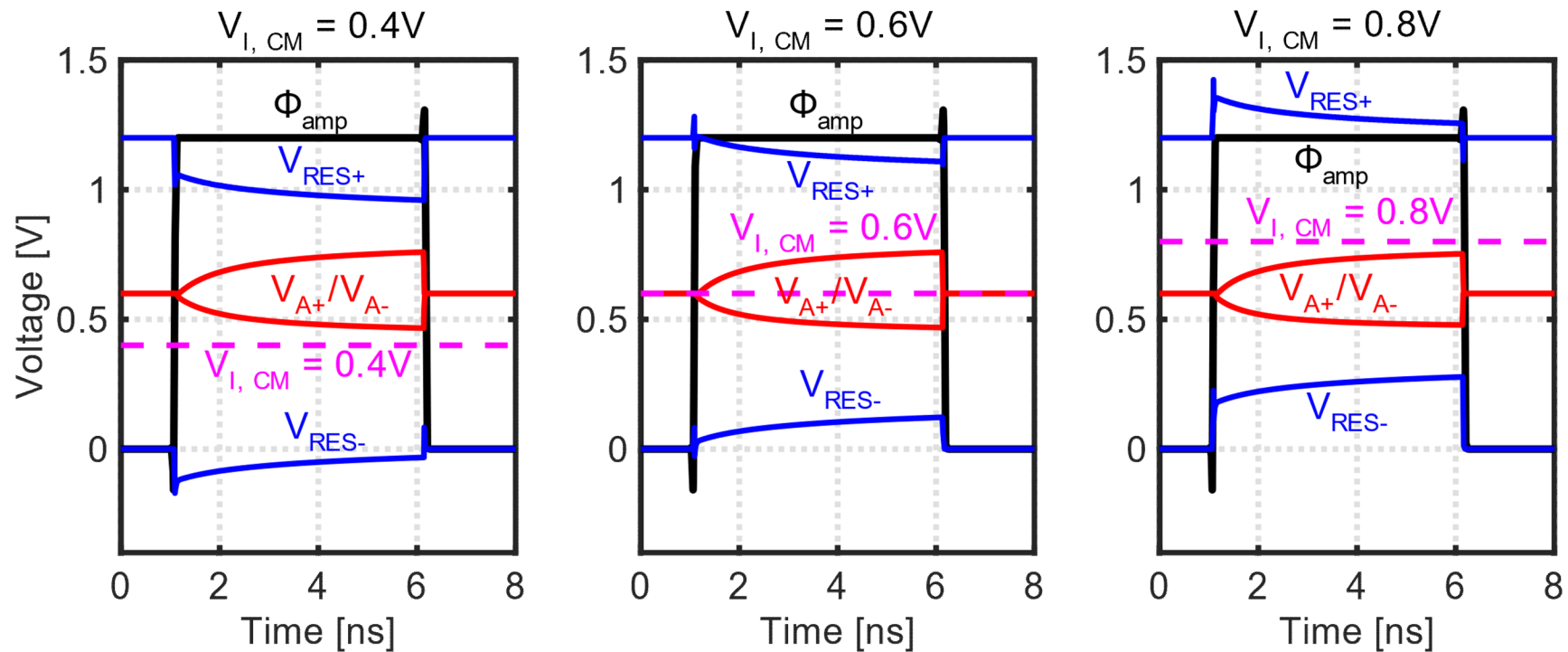
- Isolated voltage domain for FIA
 - $I_{RES+} = I_{RES-}$
 - $I_{INT+} = I_{INT-}$
 - Constant output CM voltage
- **Input CM variation**
 - Initial V_{OV} remains **constant**
 - Current reuse preserved
 - FIA operation isolated from input CM
 - CM insensitive offset/noise/power/speed



Operation of FIA

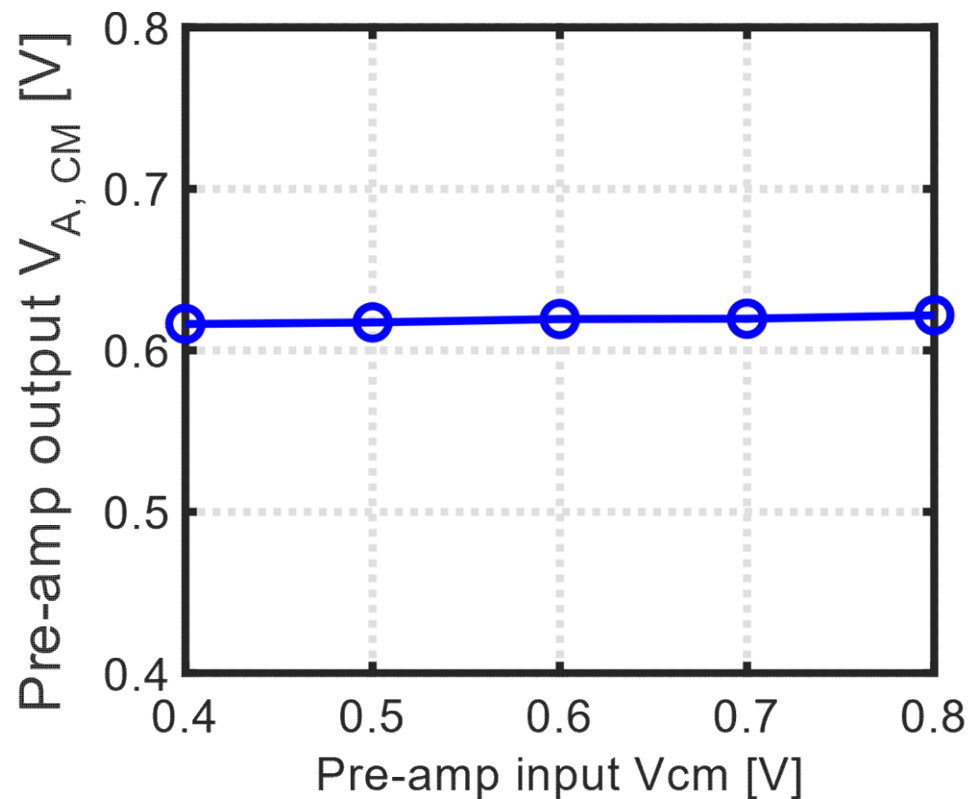
Input Common-Mode Insensitivity

- Simulated FIA behavior with input CM voltage



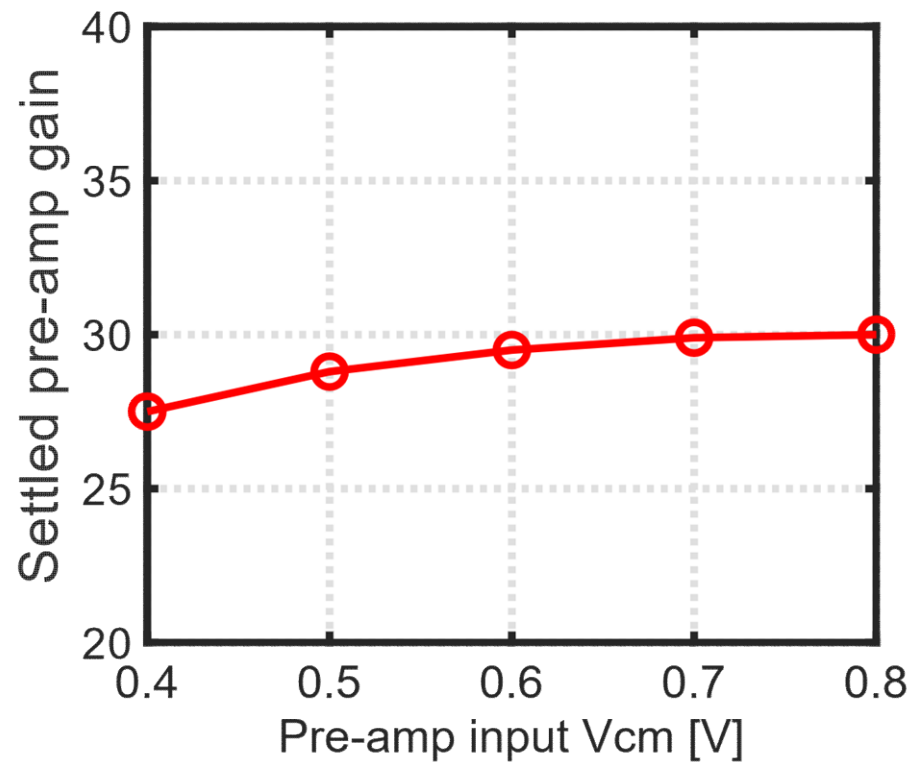
Input Common-Mode Insensitivity

- Isolated voltage domain for FIA
 - Output CM remain constant with input CM voltage varying from 0.4V to 0.8V



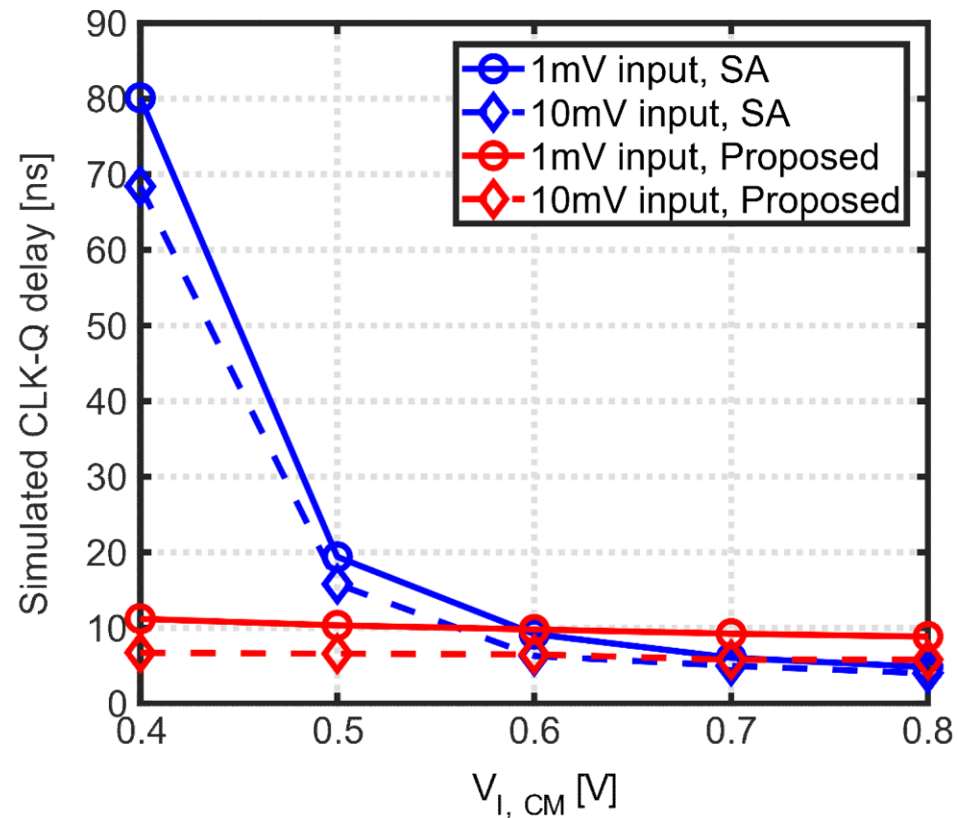
Input Common-Mode Insensitivity

- Isolated voltage domain for FIA
 - Settled FIA gain variation <10% with input CM varying from 0.4V to 0.8V



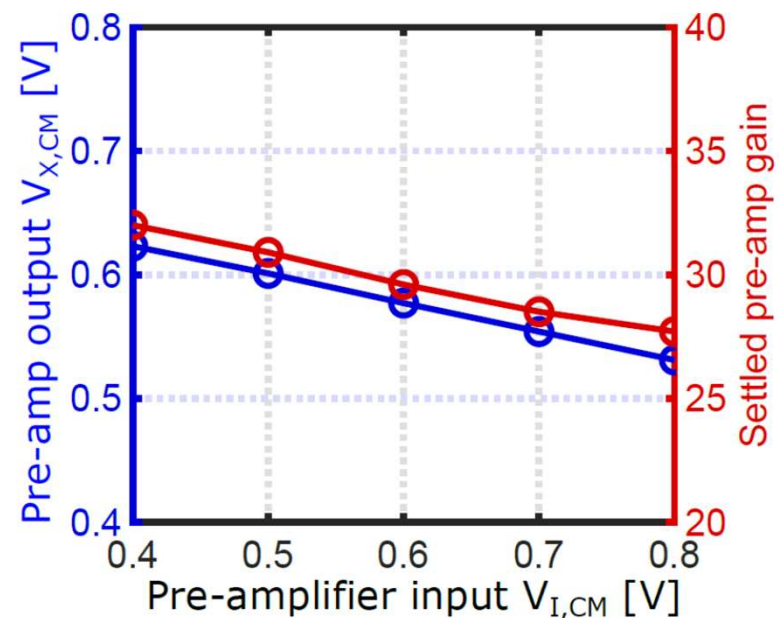
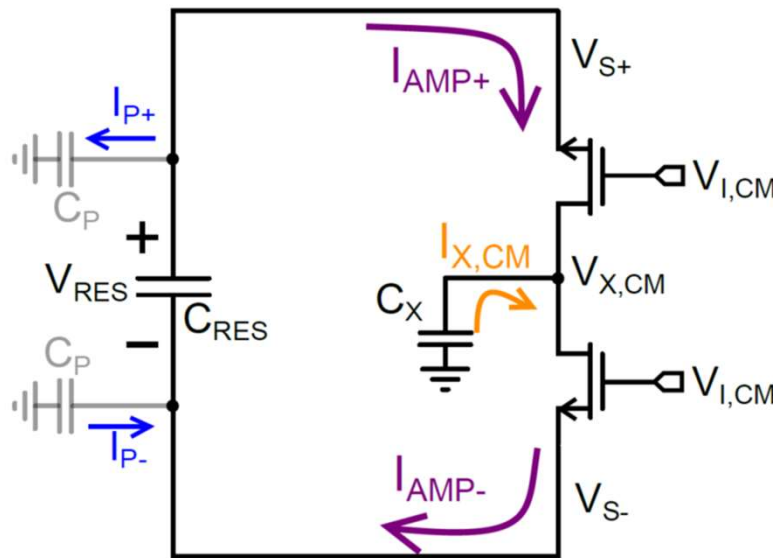
Simulated CLK-Q delay

- Simulated CLK-Q delay vs. input CM voltage
 - Strong-Arm: large delay at lower CM
 - Proposed comparator: much less delay variation



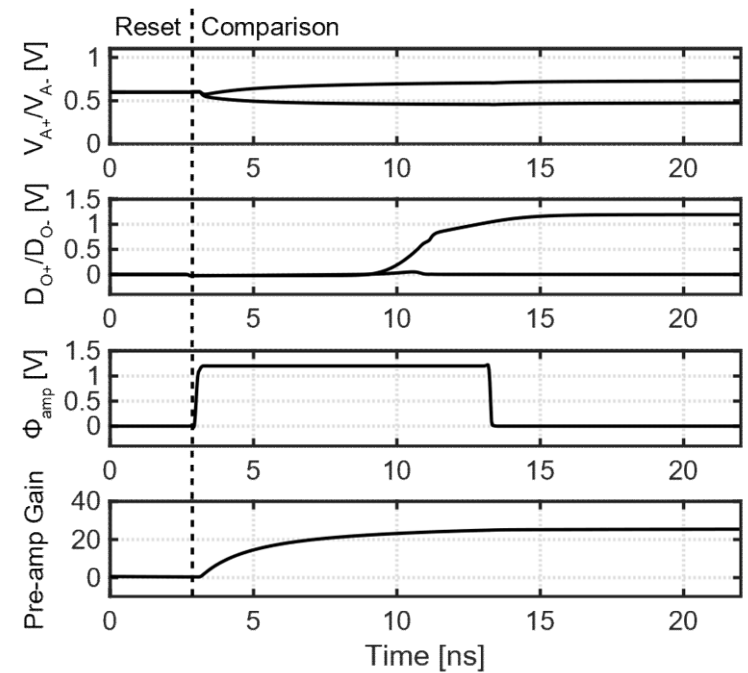
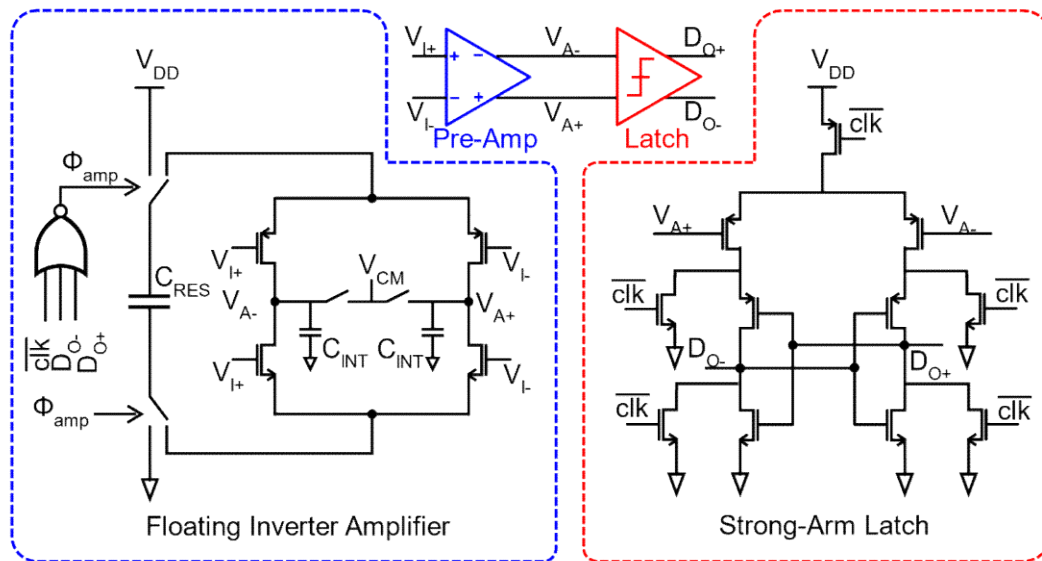
Parasitic Capacitance

- 1.5% parasitic capacitor on both sides
 - $V_{x,cm}$ variation less than 100mV for 400mV $V_{I,cm}$ variation
 - Still achieve over 30-dB improvement in CMRR



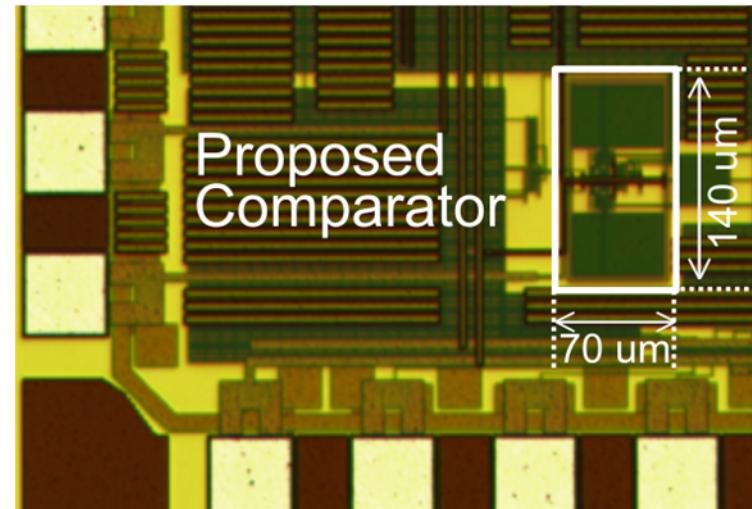
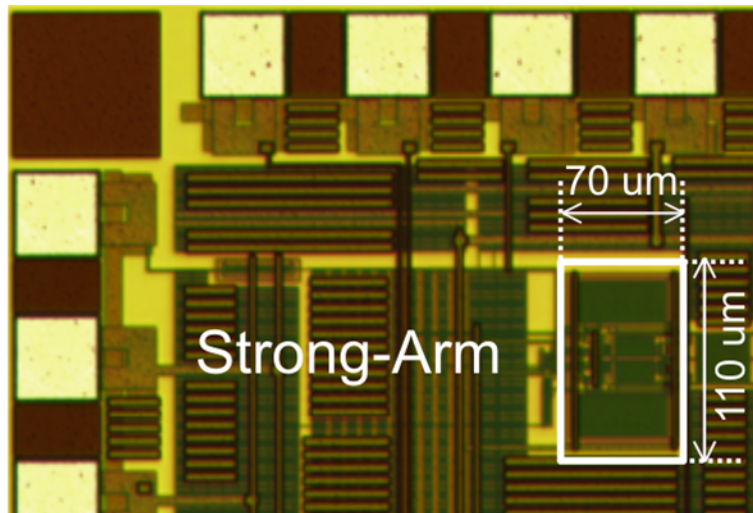
Comparator Operation

- Floating Inverter Pre-Amplifier (FIA) + Strong-Arm Latch



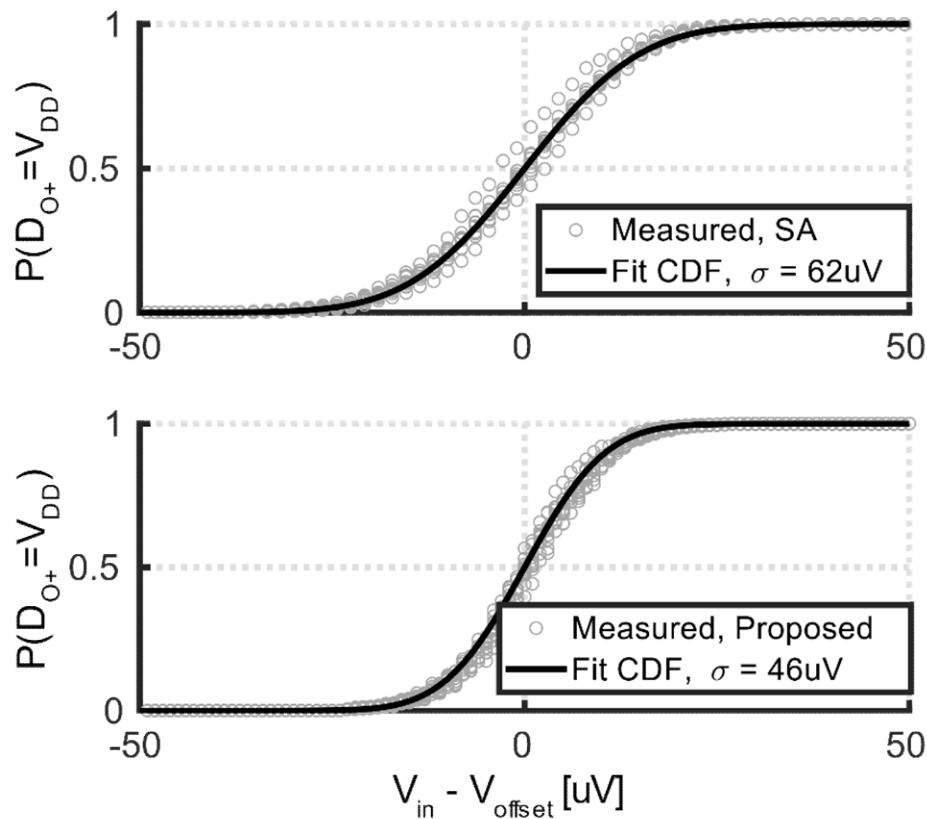
Chip Micrograph

- Fabricated in 180nm CMOS with 1.2V power supply
 - Proposed comparator
 - Standard SA latch with 2x input pair size for the same initial g_m



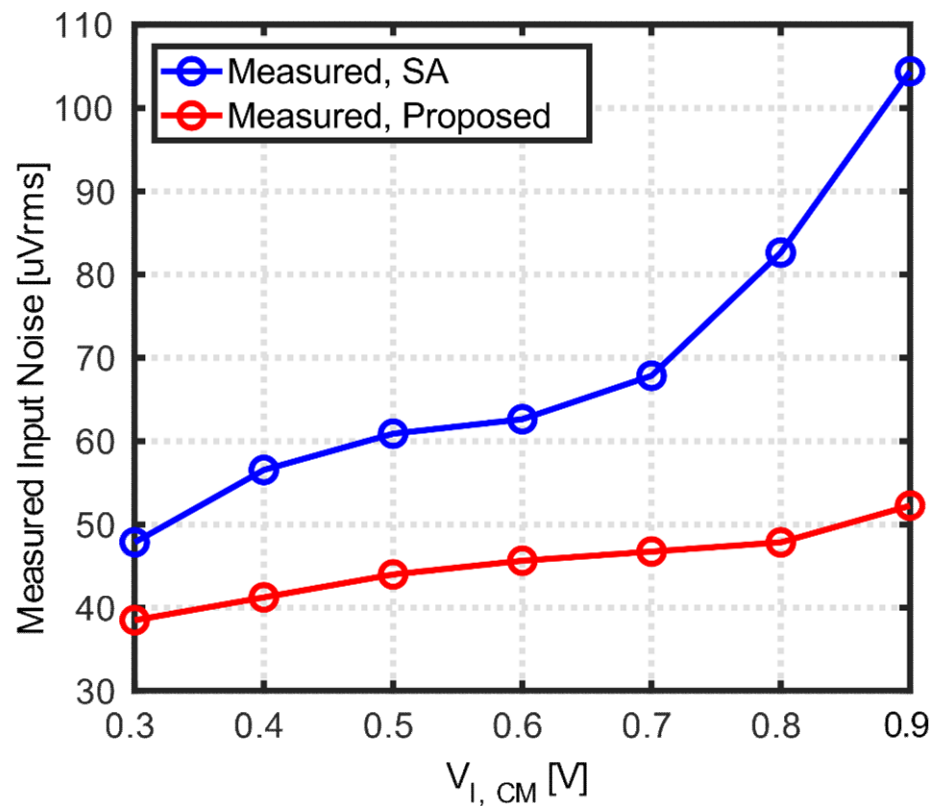
Measured Cumulative Distribution Functions (CDF)

- Measured CDF with 1.2V supply and 0.6V input CM voltage
 - Strong-Arm: 62 μV @ 4.1 pJ/conv
 - Proposed Comparator: 46 μV @ 0.98 pJ/conv



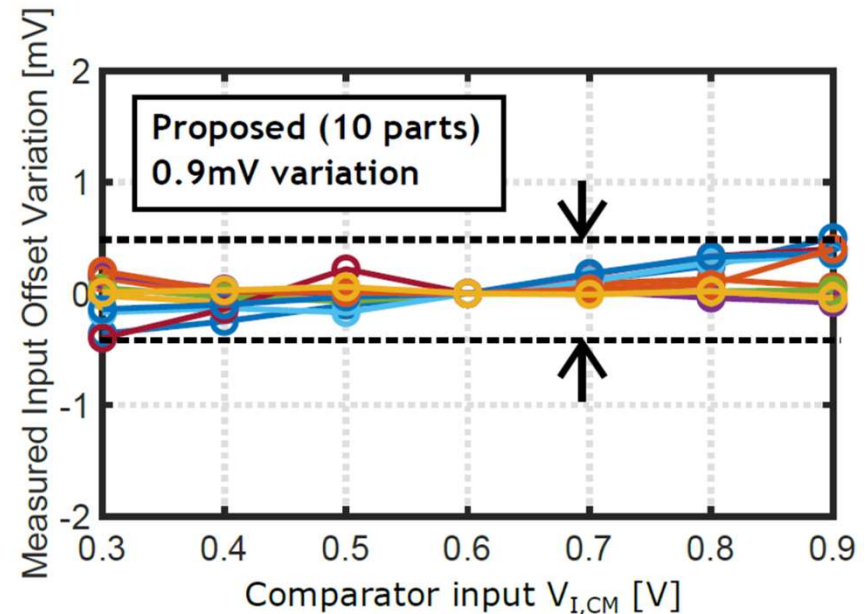
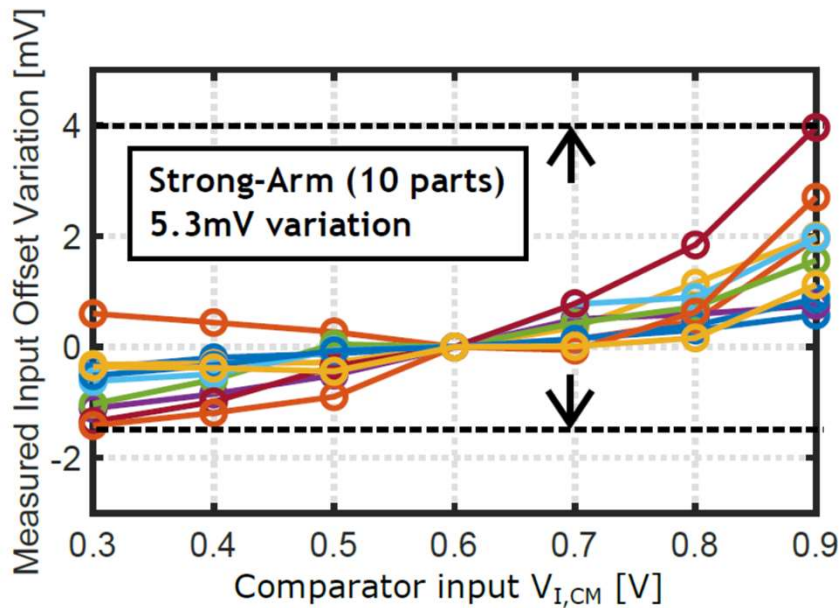
Measured Input-referred Noise

- Measured comparator noise variation vs. input CM voltage
 - Strong-Arm: large noise at higher CM
 - Proposed comparator: >4X noise variation reduction



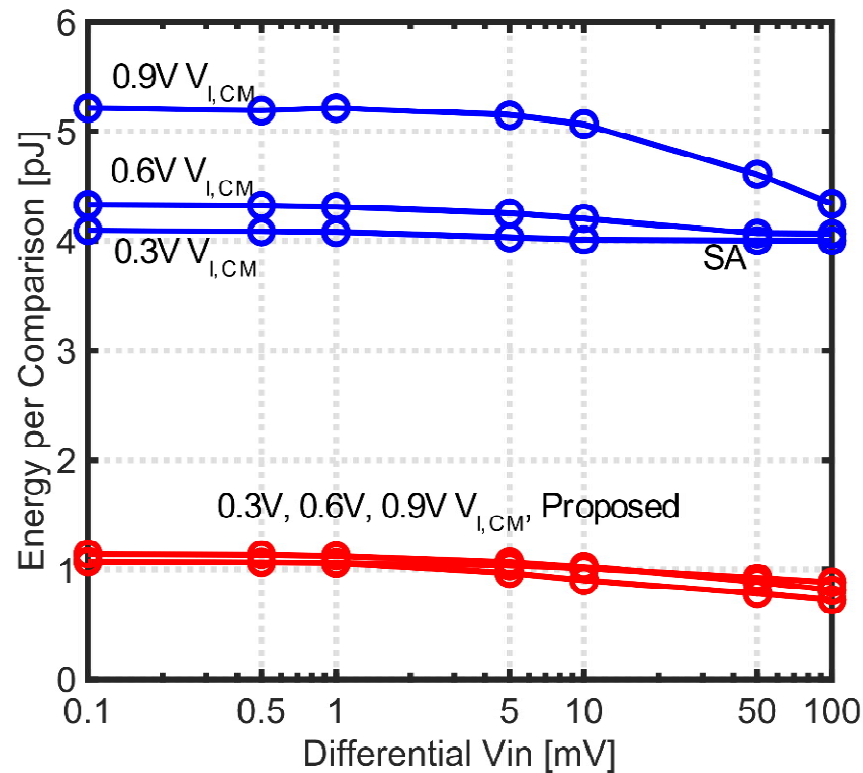
Measured Input-referred Offset

- Measured comparator offset variation vs. input CM voltage
 - Both comparators calibrated at 0.6V input CM voltage
 - Proposed comparator: >5X offset variation reduction



Measured Energy Consumption

- Measured energy consumption vs. differential input voltage
 - Proposed comparator: >4X energy reduction over all cases



Comparison with Prior Works

Specifications	This Work		[2]		[5]
Featured Architecture	Proposed FIA+SA	Standard SA	Dynamic Bias	Double-Tail [4]	Double-Tail
Process [nm]	180	180	65	65	90
Supply [V]	1.2	1.2	1.2	1.2	1.2
Noise [μ V]	46	62	400	450	1500
Energy [pJ]	0.98	4.1	0.034	0.088	0.113
FoM* [$\text{nJ} \cdot \mu\text{V}^2$]	2.07	15.8	5.44	17.8	254
Insensitive to Input CM voltage	Yes	No	No	No	No

* $FoM = \text{Energy} \cdot (\text{NoisePower})$

Objective

	Energy-efficiency				Robustness
	Current reuse	Partially discharge	Dynamic bias	Pre-amp gain	CM/Process insensitivity
Strong-Arm	☹	☹	☹	☹	☹
2-stage v1	☹	☹	☹	☹☹	☹
2-stage v2	☹	☹	☹	😊	☹
2-stage v3	☹	☹	☹	😊	☹
Dynamic Bias Int.	☹	😊	😊	😊	☹
Bidirectional Int.	😊	☹	☹	☹	☹
Cascade Input	☹	☹	☹	😊😊	☹
Objective	😊	😊	😊	😊😊😊	😊